INCH-POUND

MIL-M-28787/347(SH)
26 October 1990

MILITARY SPECIFICATION

MODULES, DIGITAL, STANDARD ELECTRONIC

CLOCK, MICROPROCESSOR MODULE, KEY CODE TRU

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic microprocessor clock module. This module is class I as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification, standards and handbook. The following specification, standards and handbook form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

HANDBOOK

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a failure rate predicted in accordance with MIL-STD-1389. The predicted failure rate shall not exceed 0.8347 failures per million operating hours when predicted in accordance with MIL-HDBK-217.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be TRU.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 100 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I. Module electrical requirements

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
A1	Isolation, initial	II	S	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
A2-A13	Isolation, initial	II	X	10	INF	10	INF	10	INF	MOhm
	end-of-life	II	Q	1	INF	1	INF	1	INF	MOhm
B1-B4	Continuity	III	X	0.0	1.0	0.0	1.0	0.0	1.0	Ohm
B5-B8	Continuity	III	Q	0.0	0.1	0.0	0.1	0.0	0.1	Ohm
C	Vcc stress	IV	Q	---------Module shall pass all remaining tests---------						
D1-D22	Ibd	V	S	0.0	100.0	0.0	100.0	0.0	100.0	uA
D23-D25	Ibd	V	S	0.0	1.0	0.0	1.0	0.0	1.0	mA
D26-D30	Ibd	V	S	0.0	50.0	0.0	50.0	0.0	50.0	uA
D31-D36	Ibd	V	S	0.0	150.0	0.0	150.0	0.0	150.0	uA
D37, D38	Ibd	V	S	0.0	300.0	0.0	300.0	0.0	300.0	uA
D39	Ibd	V	S	0.0	200.0	0.0	200.0	0.0	200.0	uA
D40	Ibd	V	S	0.0	8.4	0.0	8.4	0.0	8.4	mA
D41	Ibd	V	S	0.0	7.4	0.0	7.4	0.0	7.4	mA
E1-E22	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E23-E25	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V
E26-E39	Vic	VI	S	-1.5	-0.4	-1.5	-0.4	-1.5	-0.4	V
E40, E41	Vic	VI	S	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	V
F1-F24	Ios	VII	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
F25-F30	Ios	VII	S	-225.0	-50.0	-225.0	-50.0	-225.0	-50.0	mA
F31-F34	Ios	VII	S	-90.0	-10.0	-90.0	-10.0	-90.0	-10.0	mA
F35, F36	Ios	VII	S	-110.0	-40.0	-110.0	-40.0	-110.0	-40.0	mA
F37, F38	Ios	VII	S	-130.0	-15.0	-130.0	-15.0	-130.0	-15.0	mA
F39	Ios	VII	S	-225.8	-50.0	-225.8	-50.0	-225.8	-50.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
G1-G18	Iih	VIII	X	0.0	20.0	0.0	20.0	0.0	20.0	uA
G19-G21	Iih	VIII	X	-295.0	-247.0	-295 0	-247.0	-295.0	-247.0	uA
G22	Iih	VIII	X	0.0	20.0	0.0	20.0	0.0	20.0	uA
G23-G30	Iih	VIII	X	0.0	50.0	0.0	50.0	0.0	50.0	uA
G31-G36	Iih	VIII	X	0.0	70.0	0 0	70.0	0.0	70.0	uA
G37,G38	Iih	VIII	X	0.0	60.0	0.0	60.0	0.0	60.0	uA
G39	Iih	VIII	X	0.0	40.0	0 0	40.0	0.0	40.0	uA
G40	Iih	VIII	X	-2 95	-1 99	-2 95	-1 99	-2.95	-1.99	mA
G41	Iih	VIII	X	-2.95	-1.89	-2.95	-1.89	-2.95	-1.89	mA
H1-H18	Iil	IX	X	-400.0	0.0	-400.0	0.0	-400.0	0.0	uA
H19-H21	Iil	IX	X	-917.0	-516.0	-917 0	-516.0	-917.0	-516.0	uA
H22	Iil	IX	X	-380.0	-30.0	-380.0	-30.0	-380.0	-30.0	uA
H23-H25	Iil	IX	X	-400.0	0.0	-400 0	0.0	-400.0	0.0	uA
H26-H30	Iil	IX	X	-250.0	-1.0	-250 0	-1.0	-250.0	-1.0	uA
H31-H36	Iil	IX	X	-650.0	-1.0	-650.0	-1.0	-650.0	-1.0	uA
H37	Iil	IX	X	-1260.0	0.0	-1260 0	0.0	-1260.0	0.0	uA
H38	Iil	IX	X	-1200.0	0 0	-1200 0	0 0	-1200 0	0.0	uA
H39	Iil	IX	X	-630.0	0 0	-630.0	0.0	-630.0	0.0	uA
H40	Iil	IX	X	-40.9	-11.0	-40.9	-11.0	-40.9	-11.0	mA
H41	Iil	IX	X	-37.3	-9.8	-37.3	-9.8	-37 3	-9.8	mA
J1-J34	Voh	X	X	2.4	4.5	2 4	4.5	2.4	4.5	V
J35-J38	Voh	X	X	2.5	4.5	2 5	4.5	2.5	4.5	V
J39,J40	Voh	X	X	2.4	4.5	2 4	4.5	2 4	4.5	V
J41	Voh	X	X	2.4	2.6	2.4	2.6	2.4	2.6	V
K1-K24	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K25-K30	Vol	XI	X	0.0	0 55	0 0	0.55	0 0	0.55	V
K31-K36	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K37,K38	Vol	XI	X	0.0	0.4	0.0	0.4	0.0	0.4	V
K39	Vol	XI	X	0.0	0.55	0.0	0.55	0.0	0.55	V
K40	Vol	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
L1-L24	Iozh	XII	X	-20	20	-20	20	-20	20	uA
L25-L28	Iozh	XII	X	-100	100	-100	100	-100	100	uA
M1-M24	Iozl	XIII	X	-20	20	-20	20	-20	20	uA
M25-M28	Iozl	XIII	X	-100	100	-100	100	-100	100	uA
N	Icc	XIV	X	30	644	30	644	30	644	mA
P1,P2	Functional	XV	X	----------See table XV----------						
Q1,Q2	Dynamic	XVI	S	----------See table XVI----------						
Q3,Q4	Dynamic	XVI	Q	----------See table XVI----------						
R	PSFC	XVIII	S	3.9	7.3	3.9	7.3	3.9	7.3	uF
S	PSTR	XIX	Q	0	500	0	500	0	500	mV
T1-T24	Output crosstalk (high level)	XX	Q	0	800	0	800	0	800	mV
T25-T34	Output crosstalk (high level)	XX	Q	0	500	0	500	0	500	mV
T35,T36	Output crosstalk (high level)	XX	Q	0	400	0	400	0	400	mV

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		80 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
T37	Output crosstalk (high level)	XX	Q	0	500	0	500	0	500	mV
T38	Output crosstalk (high level)	XX	Q	0	400	0	400	0	400	mV
T39,T40	Output crosstalk (high level)	XX	Q	0	500	0	500	0	500	mV
U1-U24	Output crosstalk (low level)	XXI	Q	0	800	0	800	0	800	mV
U25-U34	Output crosstalk (low level)	XXI	Q	0	500	0	500	0	500	mV
U35,U36	Output crosstalk (low level)	XXI	Q	0	400	0	400	0	400	mV
U37,U38	Output crosstalk (low level)	XXI	Q	0	500	0	500	0	500	mV
V1-V48	Output crosstalk duration	XXII	Q	0	55	0	55	0	55	nS
W1-W6	Cin	XXIII	Q	2	40	2	40	2	40	pF
W7,W8	Cin	XXIII	Q	2	25	2	25	2	25	pF
W9	Cin	XXIII	Q	2	40	2	40	2	40	pF

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
W10-W13	Cin	XXIII	Q	2	25	2	25	2	25	pF
W14-W22	Cin	XXIII	Q	2	40	2	40	2	40	pF
W23	Cin	XXIII	Q	2	25	2	25	2	25	pF
W24-W27	Cin	XXIII	Q	2	40	2	40	2	40	pF
W28	Cin	XXIII	Q	2	30	2	30	2	30	pF
W29, W30	Cin	XXIII	Q	2	40	2	40	2	40	pF
W31-W36	Cin	XXIII	Q	4	60	4	60	4	60	pF
W37, W38	Cin	XXIII	Q	6	60	6	60	6	60	pF
W39	Cin	XXIII	Q	2	20	2	20	2	20	pF
W40	Cin	XXIII	Q	20	150	20	150	20	150	pF
W41	Cin	XXIII	Q	30	160	30	160	30	160	pF
X1-X24	Cout	XXIV	Q	2	50	2	50	2	50	pF
X25-X28	Cout	XXIV	Q	2	35	2	35	2	35	pF
Y	Threshold	XXV	X	See table XXV						
Z1-Z24	Disable time	XXVI	Q	8	39	8	39	8	39	nS
Z25-Z48	Disable time	XXVI	Q	8	73	8	73	8	73	nS
Z49-Z52	Disable time	XXVI	Q	2	40	2	40	2	40	nS
Z53-Z56	Disable time	XXVI	Q	2	51	2	51	2	51	nS
AA1, AA2	Output pulse width and pulse period	XXVII	X	117.0	133.0	117.0	133.0	117.0	133.0	nS
AA3	Output pulse width and pulse period	XXVII	X	249.98	250.02	249.98	250.02	249.98	250.02	nS

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		0 ± 5°C 4/		60 ± 5°C 4/		Units
				Min	Max	Min	Max	Min	Max	
AA4,AA5	Output pulse width and pulse period	XXVII	X	78.0	88.0	78.0	88.0	78.0	88.0	nS
AA4,AA5	Output pulse width and pulse period	XXVII	X	78.0	88.0	78 0	88.0	78.0	88.0	nS
AA6	Output pulse width and pulse period	XXVII	X	166.65	166 68	166.65	166.68	166.65	166.68	nS
AA7,AA8	Output pulse width and pulse period	XXVII	X	78.0	88.0	78.0	88.0	78.0	88.0	nS
AA9	Output pulse width and pulse period	XXVII	X	166.65	166.68	166 65	166.68	166.65	166.68	nS
AA10,AA11	Output pulse width and pulse period	XXVII	X	35.0	48.0	35 0	48.0	35.0	48.0	nS
AA12	Ouput pulse width and pulse period	XXVII	X	83.32	83.34	83.32	83.34	83.32	83.34	nS
AA13,AA14	Output pulse width and pulse period	XXVII	X	35.0	48.0	35.0	48.0	35.0	48.0	nS
AA15	Output pulse width and pulse period	XXVII	X	83.32	83 34	83 32	83.34	83.32	83.34	nS

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/347-1. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. The module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.0 $\pm$ 0.5 V dc, except the output short circuit current limits apply only for a power supply voltage of 5.5 V dc $\pm$ 1 percent.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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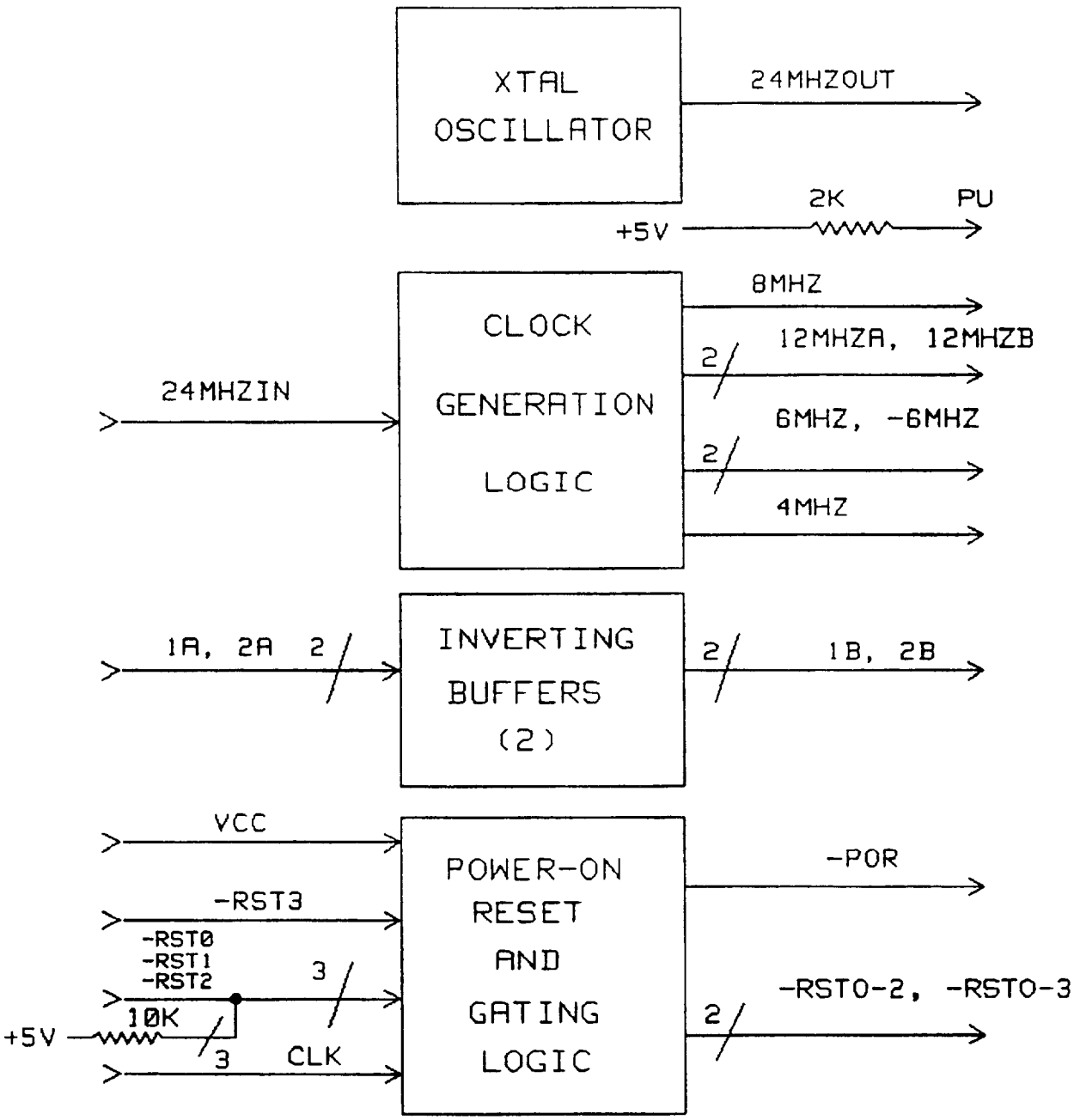


FIGURE 1. TRU functional block diagram.

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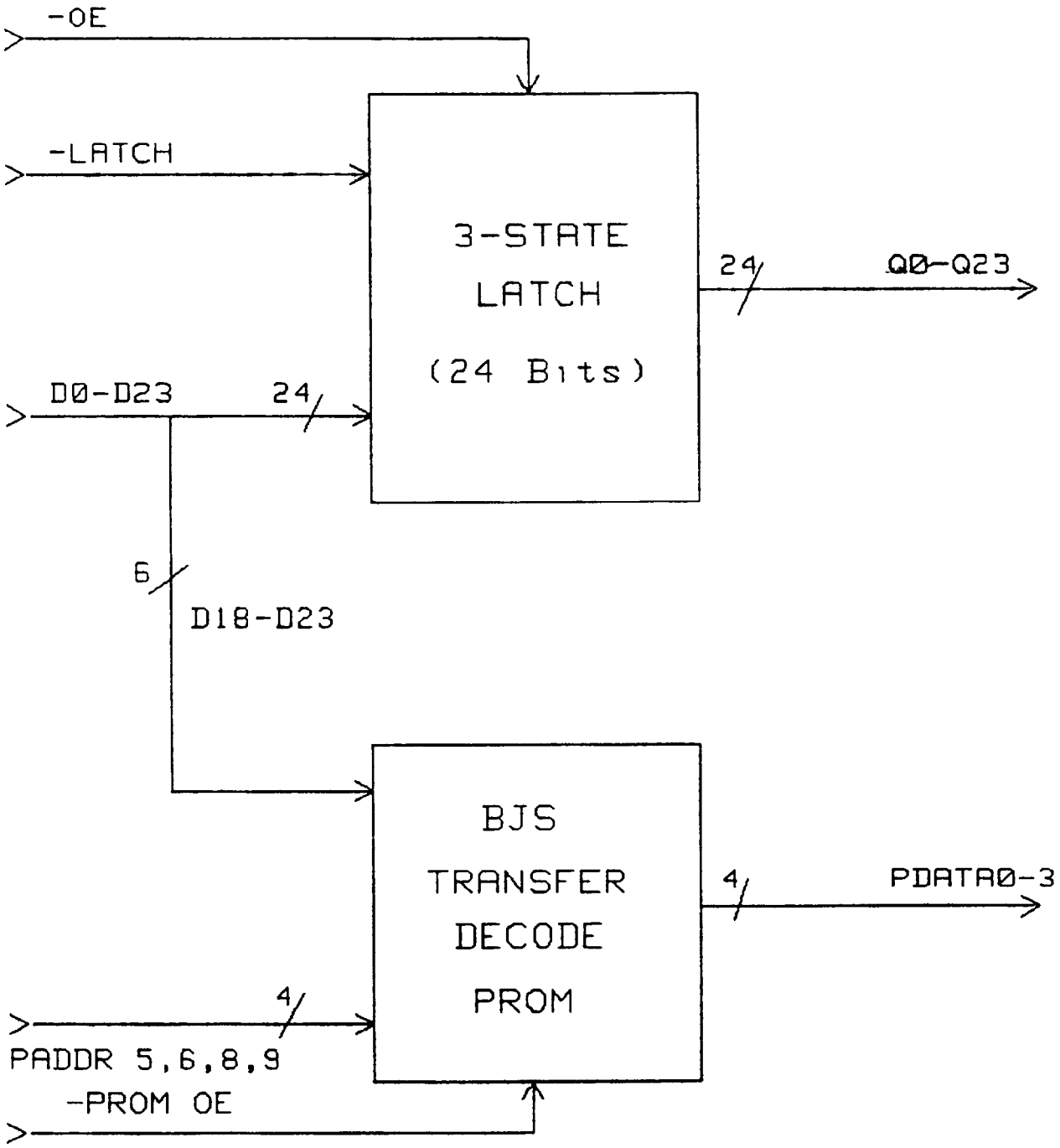


FIGURE 1. TRU functional block diagram - Continued.

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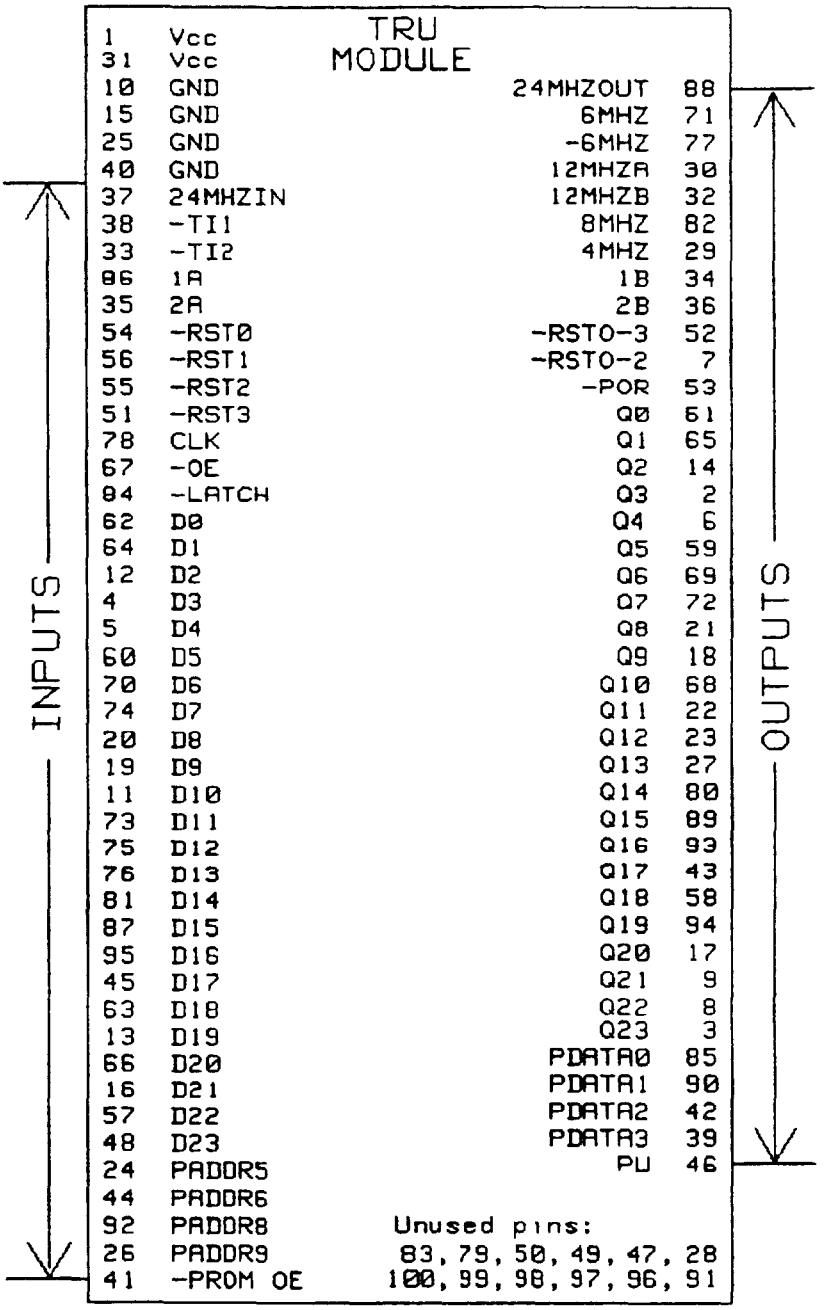


FIGURE 2. TRU pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through AA shall be performed in any order with the following exceptions. Test conditions A, C, D, E and F are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or F are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E and F where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

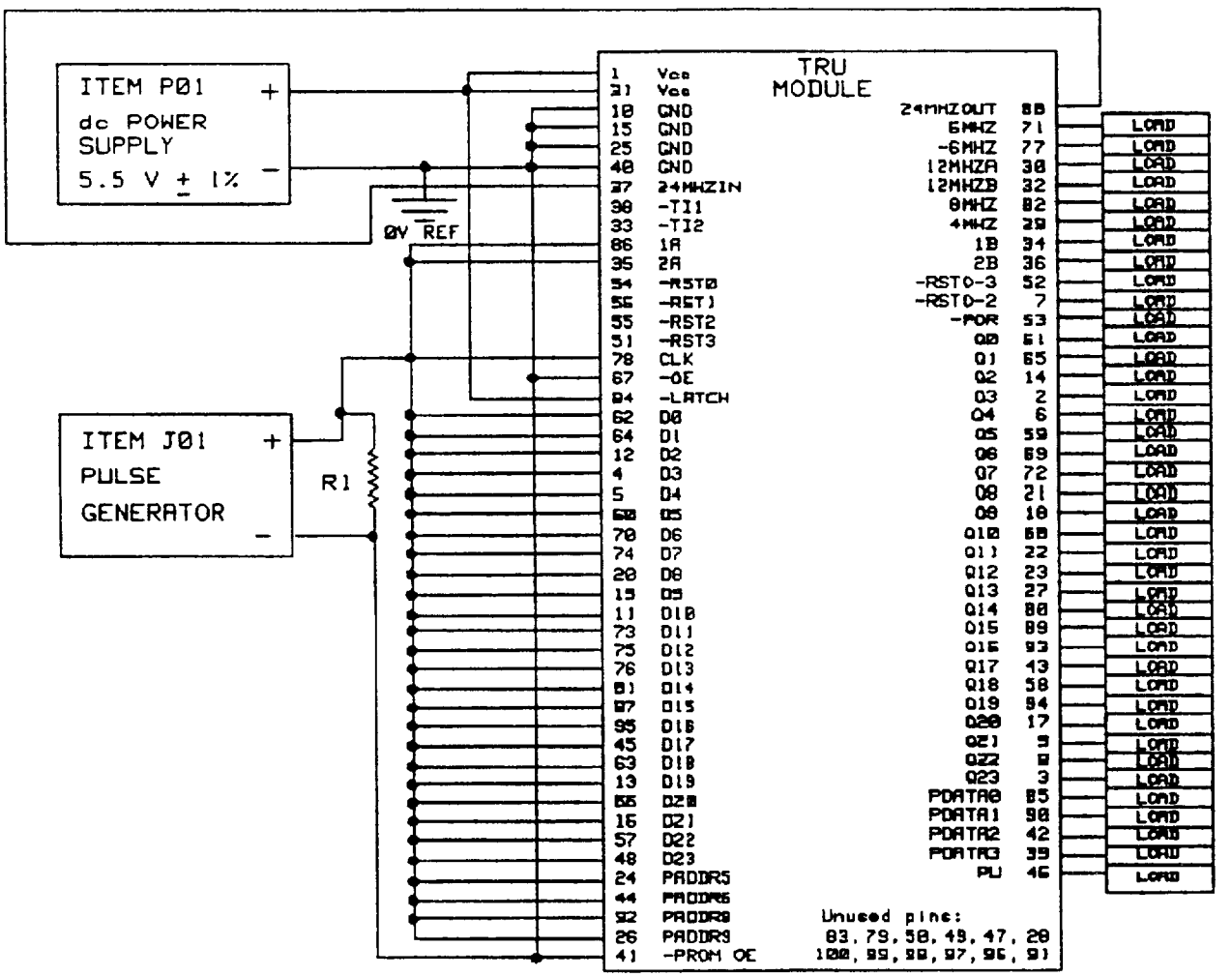
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through AA in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, G through R, Y and AA shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated as shown on figures 3 and 4.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 6. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition R, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTES:

1. Load outputs as specified in the output voltage test.
2. R1 = 51 Ohms ± 5 percent, 1 W.

FIGURE 3. TRU operating temperature and life test circuit.

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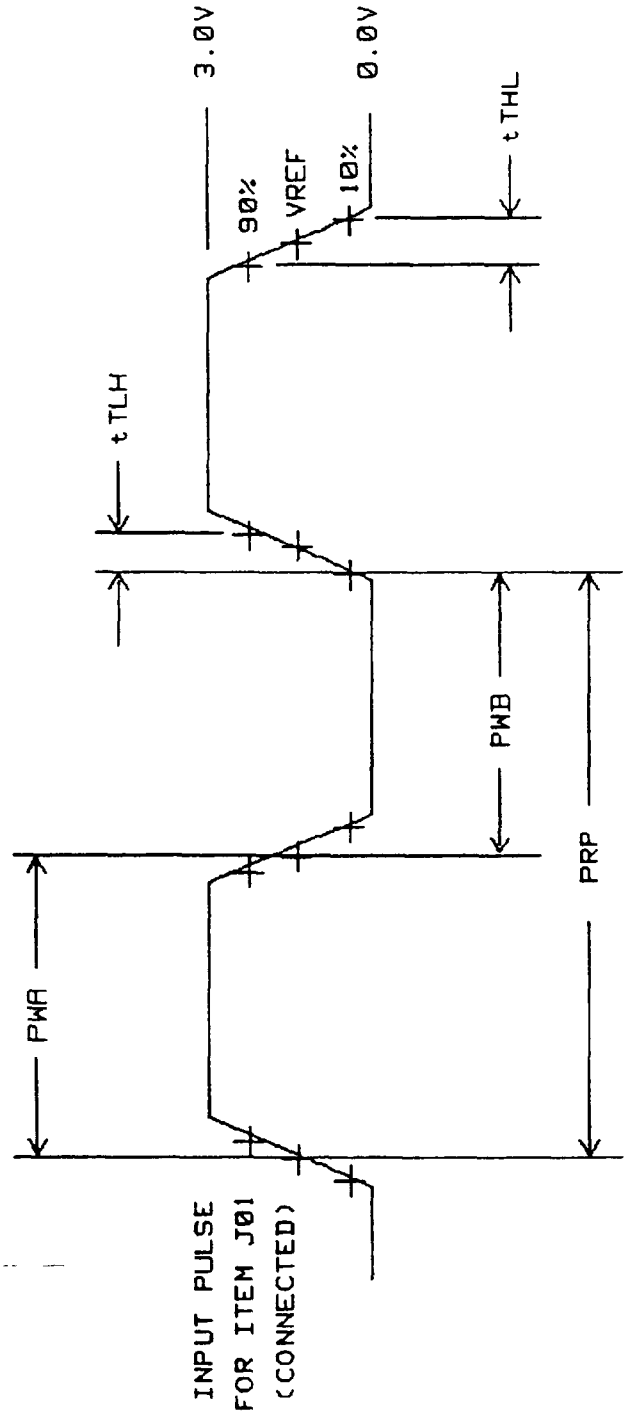


FIGURE 4. Operating temperature and life test waveforms.

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NOTE:

1. Input pulse parameters for item J01:

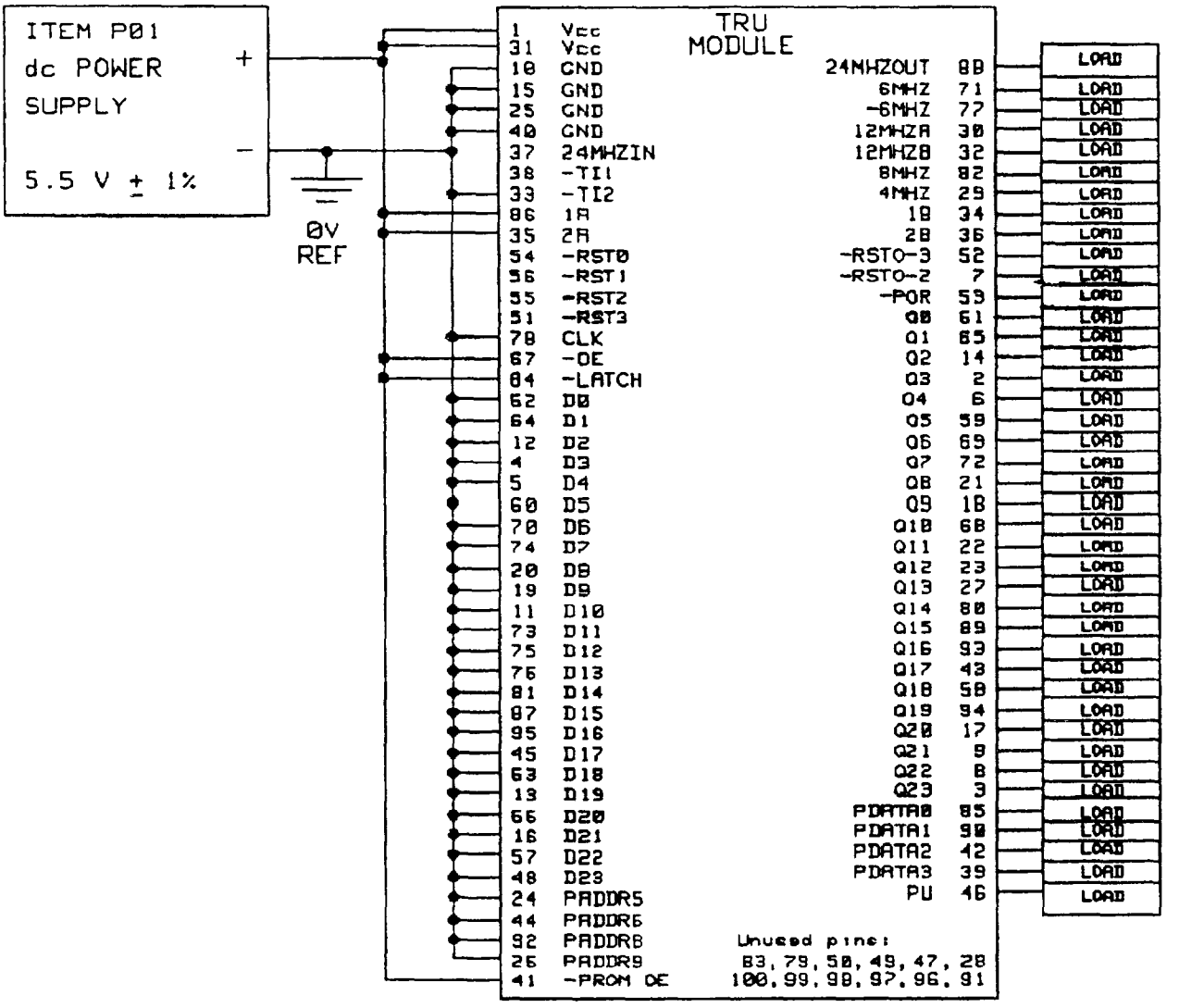
   PWA = 250 ± 25 nS.

   PWB = 250 ± 25 nS.

   PRP = 500 ± 50 nS.

   tTLH = tTHL = 50 ± 10 nS.

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NOTE:

1. Load outputs as specified in the output voltage test.

FIGURE 5. TRU component temperature test circuit.

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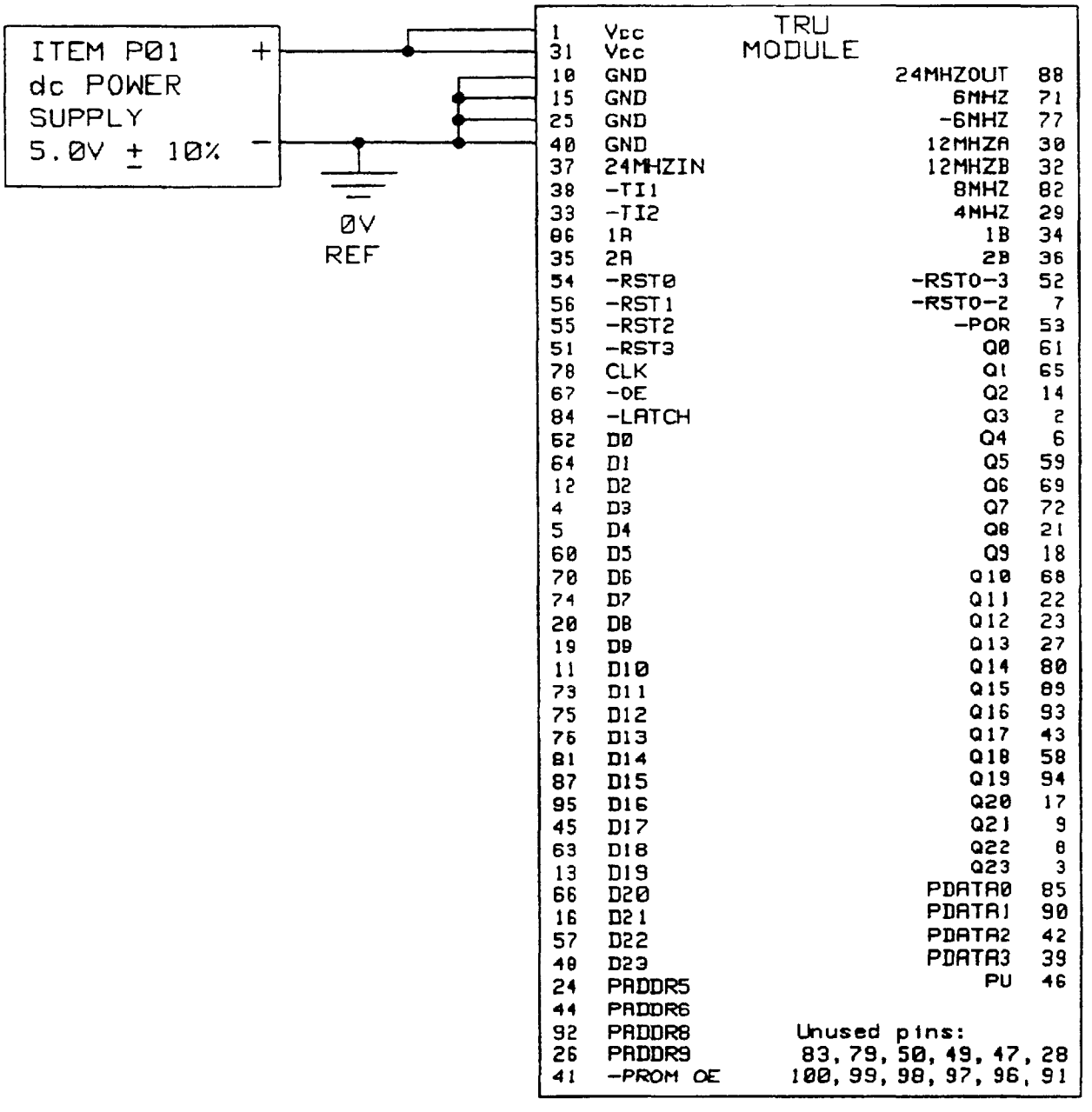


FIGURE 6. TRU vibration test circuit.

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4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figures 3 and 4. Following the life test and while still at the maximum class temperature, electrical tests B, G through R, Y and AA shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, G through R, Y and AA shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Cout = output capacitance.
(c) Ibd = input breakdown current.
(d) Icc = power supply current.
(e) Iih = input high level current.
(f) Iil = input low level current.
(g) Iin = input forcing current.
(h) Ioh = output high level current.
(i) Iol = output low level current.
(j) Ios = output short circuit current.
(k) Iozh = output high level leakage current.
(l) Iozl = output low level leakage current.
(m) NA = not applicable.
(n) PSFC = power supply filter capacitance.
(o) PSTA = power supply transient amplitude.
(p) Vic = input clamp diode voltage.
(q) Vih = input high level voltage.
(r) Vil = input low level voltage.
(s) Vin = input forcing voltage.
(t) Voh = output high level voltage.
(u) Vol = output low level voltage.
(v) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 3.542 W (max); 1.878 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
D0	-1.5	0.7	2.0	-0.4	20	100	40
D1	-1.5	0.7	2.0	-0.4	20	100	40
D2	-1.5	0.7	2.0	-0.4	20	100	40
D3	-1.5	0.7	2.0	-0.4	20	100	40
D4	-1.5	0.7	2.0	-0.4	20	100	40
D5	-1.5	0.7	2.0	-0.4	20	100	40
D6	-1.5	0.7	2.0	-0.4	20	100	40
D7	-1.5	0.7	2.0	-0.4	20	100	40
D8	-1.5	0.7	2.0	-0.4	20	100	40
D9	-1.5	0.7	2.0	-0.4	20	100	40
D10	-1.5	0.7	2.0	-0.4	20	100	40
D11	-1.5	0.7	2.0	-0.4	20	100	40
D12	-1.5	0.7	2.0	-0.4	20	100	40

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
D13	-1.5	0.7	2.0	-0.4	20	100	40
D14	-1.5	0.7	2.0	-0.4	20	100	40
D15	-1.5	0.7	2.0	-0.4	20	100	40
D16	-1.5	0.7	2.0	-0.4	20	100	40
D17	-1.5	0.7	2.0	-0.4	20	100	40
-RST0	-1.5	0.7	2.0	-0.92	-295	100	25
-RST1	-1.5	0.7	2.0	-0.92	-295	100	25
-RST2	-1.5	0.7	2.0	-0.92	-295	100	25
-RST3	-1.5	0.7	2.0	-0.38	20	100	25
-OE	-1.5	0.7	2.0	-1.26	60	300	60
-LATCH	-1.5	0.7	2.0	-1.2	60	300	60
CLK	-1.5	0.7	2.0	-0.63	40	200	20
PADDR5	-1.5	0.8	2.0	-0.25	50	50	40
PADDR6	-1.5	0.8	2.0	-0.25	50	50	40

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (uA)	Ibd max (uA)	Cin max (pF)
PADDR8	-1.5	0.8	2.0	-0.25	50	50	40
PADDR9	-1.5	0.8	2.0	-0.25	50	50	40
24MHZIN	-1.5	0.8	2.0	-0.40	50	1000	25
1A	-1.2	0.8	2.0	-0.40	50	1000	25
2A	-1.2	0.8	2.0	-0.40	50	1000	25
D18	-1.5	0.7	2.0	-0.61	70	150	60
D19	-1.5	0.7	2.0	-0.61	70	150	60
D20	-1.5	0.7	2.0	-0.61	70	150	60
D21	-1.5	0.7	2.0	-0.61	70	150	60
D22	-1.5	0.7	2.0	-0.61	70	150	60
D23	-1.5	0.7	2.0	-0.61	70	150	60
-PROM OE	-1.5	0.8	2.0	-0.25	50	50	30
-TI1	-1.2	0.8	2.0	-40.9	-2950	8400	150
-TI2	-1.2	0.8	2.0	-37.3	-2950	7400	160

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh min (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
Q0	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q1	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q2	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q3	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q4	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q5	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q6	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q7	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q8	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q9	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q10	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q11	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q12	2.4	0.4	-1.0	12.0	20	-20	-15	50

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
Q13	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q14	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q15	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q16	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q17	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q18	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q19	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q20	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q21	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q22	2.4	0.4	-1.0	12.0	20	-20	-15	50
Q23	2.4	0.4	-1.0	12.0	20	-20	-15	50
12MHZA	2.4	0.55	-3.0	48.0	NA	NA	-50	NA
12MHZB	2.4	0.55	-3.0	48.0	NA	NA	-50	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
8MHZ	2.4	0.55	-3.0	48.0	NA	NA	-50	NA
4MHZ	2.4	0.55	-3.0	48.0	NA	NA	-50	NA
1B	2.4	0.55	-3.0	48.0	NA	NA	-50	NA
2B	2.4	0.55	-3.0	48.0	NA	NA	-50	NA
PDATA0	2.4	0.5	-2.0	12.0	100	-100	-10	35
PDATA1	2.4	0.5	-2.0	12.0	100	-100	-10	35
PDATA2	2.4	0.5	-2.0	12.0	100	-100	-10	35
PDATA3	2.4	0.5	-2.0	12.0	100	-100	-10	35
6MHZ	2.5	0.5	-1.0	20.0	NA	NA	-40	NA
-6MHZ	2.5	0.5	-1.0	20.0	NA	NA	-40	NA
-RSTO-3	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
-RSTO-2	2.5	0.4	-0.4	4.0	NA	NA	-15	NA
-POR	2.4	0.55	-2.96	47.24	NA	NA	-50	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh min (mA)	Iol max (mA)	Iozh max (uA)	Iozl min (uA)	Ios max (mA)	Cout max (pF)
24MHZOUT	2.4	0.5	-0.24	9.6	NA	NA	NA	NA
PU	2.4	NA	-1.0	NA	NA	NA	NA	NA

6.3.3 Dynamic parameters. The dynamic parameters appear with the dynamic test.

6.4 Module function. The TRU module contains a 24 MHZ oscillator and clock divider circuitry. When 24MHZOUT is connected to 24MHZIN, the oscillator frequency is divided down to produce 12MHZ, 6MHZ and 4MHZ output signals (each with a 50 percent duty cycle). An 8MHZ (33 percent duty cycle) output is also provided. The connection from 24MHZOUT to 24MHZIN should be as short as possible. A power-on-reset circuit generates a reset pulse of at least 100 mS when power is initially applied to the module. The power supply rise time (0.0 V to 4.75 V dc) must be less than 230 mS in order to insure this 100 mS minimum pulse width. The power-on-reset circuit also generates a reset pulse when Vcc drops to below 1 volt for 400 mS or more. If the power supply drops to a voltage between 4.5 V and 1.0 V, one of the following conditions will exist:

(a) A full (100 mS) power-on-reset pulse will be generated.

(b) A power-on-reset pulse of less than 100 mS will be generated.

(c) No power-on-reset pulse will be generated.

The TRU module contains a pulse stretcher circuit which may be used to condition external resets to meet 68000 microprocessor requirements. When one of the reset inputs (-RST0, -RST1 or -RST2) goes low, outputs -RSTO-3 and -RSTO-2 go low for seven full cycles of the clock (CLK) input (regardless of changes in the reset input). Note that the 68000 microprocessor requires reset inputs to be active for at least 10 processor clock cycles. Since this pulse stretcher only stretches pulses for 7 clock cycles, the clock input to the pulse stretcher circuit must be somewhat slower than the microprocessor clock. A 24-bit

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transparent latch is pinned out for general purpose use. When -LATCH is high, the latches are transparent and any data appearing on inputs D0-D23 is transferred to outputs Q0-Q23, respectively. On the falling edge of -LATCH, all data on inputs D0-D23 that meets the setup time is latched. These latches may be used to hold the address lines valid later in a 68000 microprocessor bus cycle. A 1K x 4 bit PROM is included on the module for use as a signal decoder. Address inputs to the PROM are (in ascending order) D18, D19, D20, D21, D22, PADDR5, PADDR6, D23, PADDR8 and PADDR9. Note that inputs D18-D23 are also connected to the latches. The PROM outputs are PDATA0, PDATA1, PDATA2 and PDATA3. These outputs should be used to drive the PROM inputs (PROM0 - PROM3) on the ASC module (MIL-M-28787/351), but have little use otherwise. When -PROM OE is high, the PROM outputs are in the high impedance state. Below are the PROM contents by HEX address:

Hex address	Hex data	Hex address	Hex data	Hex address	Hex data	Hex address	Hex data
000-087	5	2B0-2BF	5	3B5-3B7	A	3F2	2
088-097	6	2C0-2C7	6	3B8	0	3F3	3
098-0A7	5	2C8-2CF	5	3B9	1	3F4	8
0A8-0AF	6	2D0-2E7	6	3BA	2	3F5-3F7	A
0B0-0BF	5	2E8-2F7	5	3BB	3	3F8-3FF	7
0C0-0C7	6	2F8-2FF	6	3BC	8		
0C8-0CF	5	300-37F	4	3BD-3BF	A		
0D0-0EF	6	380-387	A	3C0-3C7	7		
0E8-0F7	5	388-397	7	3C8	0		
0F8-0FF	6	398	0	3C9	1		
100-187	5	399	1	3CA	2		
188-197	6	39A	2	3CB	3		
198-1A7	5	39B	3	3CC	8		
1A8-1AF	6	39C	8	3CD-3CF	A		
1B0-1BF	5	39D	A	3D0-3DF	7		
1C0-1C7	6	39E	A	3E0-3E7	7		
1C8-1CF	5	39F	A	3E8	0		
1D0-1E7	6	3A0-3A7	A	3E9	1		
1E8-1F7	5	3A8-3AF	7	3EA	2		
1F8-1FF	6	3B0	0	3EB	3		
200-287	5	3B1	1	3EC	8		
288-297	6	3B2	2	3ED-3EF	A		
298-2A7	5	3B3	3	3F0	0		
2A8-2AF	6	3B4	8	3F1	1		

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6.4.1 Functional description of I/O pins.

Signal name	Pin number	I/O	Functional description
D0	62	I	Inputs to a 24-bit transparent latch. When -LATCH is high, all latches are transparent. On the falling edge of -LATCH, the data on D0-D23 that meets the setup time is latched on outputs Q0-Q23, respectively. D18-D23 are also decoded by the PROM.
D1	64	I	
D2	12	I	
D3	4	I	
D4	5	I	
D5	60	I	
D6	70	I	
D7	74	I	
D8	20	I	
D9	19	I	
D10	11	I	
D11	73	I	
D12	75	I	
D13	76	I	
D14	81	I	
D15	87	I	
D16	95	I	
D17	45	I	
D18	63	I	
D19	13	I	
D20	66	I	
D21	16	I	
D22	57	I	
D23	48	I	
-RST0	54	I	Active low inputs to a pulse stretcher circuit. When any input goes low, the outputs (-RST0-2 and -RST0-3) go low for seven full clock (CLK) cycles.
-RST1	56	I	
-RST2	55	I	
-RST3	51	I	An active low input which is tied to an AND gate. When -RST3 is active, the output (-RST0-3) is active (low).

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
24MHZIN	37	I	Input to the clock divider circuit. When 24MHZOUT is tied to this input, output frequencies are 12MHZ, 6MHZ and 4MHZ (each with a 50 percent duty cycle). An 8MHZ (33 percent duty cycle) output is also provided.
1A 2A	86 35	I I	Inverter inputs which correspond to outputs 1B and 2B, respectively.
PADDR5 PADDR6 PADDR8 PADDR9	24 44 92 26	I I I I	Four of the ten address inputs to a PROM. The other six address inputs are connected to D18-D23.
-PROM OE	41	I	Output enable signal for the PROM. When -PROM OE is low (active), all outputs (PDATA0-PDATA3) are enabled. When -PROM OE is high, all outputs are in the high impedance state.
-OE	67	I	Output enable signal for the 24-bit transparent latch. When -OE is active (low), Q0-Q23 (the outputs) are enabled. When -OE is high, Q0-Q23 are in the high impedance state.
-LATCH	84	I	The latch enable input signal to the 24-bit transparent latch. When -LATCH is high, the latches are transparent. On the falling edge of -LATCH, any data on inputs D0-D23 that meets the setup time is latched on outputs Q0-Q23.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
CLK	78	I	Rising edge sensitive clock input to a counter in the pulse stretcher circuit. The output pulse is active for 7 clock cycles (minimum).
-TI1	38	I	Test inputs that have no user function. These inputs should be left open.
-TI2	33	I	
Q0	61	O	Outputs of the 24-bit transparent latch. When -LATCH is high, the latches are transparent and changes on the data inputs (D0-D23) are reflected on the outputs. On the falling edge of -LATCH, the input data is latched. -OE is the output enable for these outputs. When -OE is high, all outputs are set to the high impedance state.
Q1	65	O	
Q2	14	O	
Q3	2	O	
Q4	6	O	
Q5	59	O	
Q6	69	O	
Q7	72	O	
Q8	21	O	
Q9	18	O	
Q10	68	O	
Q11	22	O	
Q12	23	O	
Q13	27	O	
Q14	80	O	
Q15	89	O	
Q16	93	O	
Q17	43	O	
Q18	58	O	
Q19	94	O	
Q20	17	O	
Q21	9	O	
Q22	8	O	
Q23	3	O	
12MHZA	30	O	12MHZ outputs which have a 50 percent (typical) output.
12MHZB	32	O	

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
4MHZ	29	O	A 4MHZ output which has a 50 percent (typical) output.
8MHZ	82	O	An 8MHZ output which, typically, has a 33 percent duty cycle.
1B	34	O	Inverter outputs which correspond to inputs 1A and 2A, respectively.
2B	36	O	
PDATA0	85	O	Outputs of a signal decode PROM. PROM address inputs are D18, D19, D20, D21, D22, PADDR5, PADDR6, D23, PADDR8 and PADDR9 (LSB to MSB). The state of each output is dependent upon the address inputs.
PDATA1	90	O	
PDATA2	42	O	
PDATA3	39	O	
6MHZ	71	O	A 6MHZ clock output which has a typical duty cycle of 50 percent.
-6MHZ	77	O	A 6MHZ clock output which has a typical duty clock of 50 percent. -6MHZ is 180 degrees (typically) out of phase with 6MHZ.
-RSTO-3	52	O	An active low output that is the logical AND of the pulse stretcher output (see -RSTO) and the power-on-reset output (see -POR).
-RSTO-2	7	O	An active low output that is the logical AND of the pulse stretcher output (see -RSTO) and the power-on-reset output (see -POR).
-POR	53	O	An output that is active (low) for at least 100 mS after power is applied to the module.

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6.4.1 Functional description of I/O pins - Continued.

Signal name	Pin number	I/O	Functional description
24MHZOUT	88	O	Output of a precision 24 MHZ, TTL compatible oscillator. This output should only be tied to 24MHZIN.
PU	46	O	A logic "1" source which may be used to pull up unused inputs.

6.5 Description of tests. The following tables and associated figures describe the tests to be performed.

(a) Isolation test - table II.

(b) Continuity test - table III.

(c) Vcc stress test - table IV.

(d) Ibd test - table V.

(e) Vic test - table VI.

(f) Ios test - table VII.

(g) Iih test - table VIII.

(h) Iil test - table IX.

(i) Voh test - table X, figure 7.

(j) Vol test - table XI, figure 7.

(k) Iozh test - table XII.

(l) Iozl test - table XIII.

(m) Icc test - table XIV.

(n) Functional test - table XV, figure 8.

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(o) Dynamic test - tables XVI and XVII, figures 9 and 10.

(p) PSFC test - table XVIII.

(q) PSTA test - table XIX, figures 8 and 11.

(r) Output crosstalk tests - tables XX and XXI, figures 7, 8, 12 and 13.

(s) Output crosstalk duration test - table XXII, figures 7 and 14.

(t) Cin test - table XXIII.

(u) Cout test - table XXIV.

(v) Threshold test - table XXV.

(w) Disable time test - table XXVI, figures 7 and 15 through 18.

(x) Output pulse width and pulse period test - table XXVII, figures 7 and 19.

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6.6 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Preparing activity:
Navy - SH

(Project 5963-N742)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A1	Isolation, frame	All pins	Frame	10	INF	MOhm	1-7
A2	Isolation, used to unused	All used pins	All unused pins	10	INF	MOhm	1-7
A3	Isolation, unused to unused	79	83,91,96, 97,98,99, 100,28, 47,49,50	10	INF	MOhm	1-7
A4	Isolation, unused to unused	83	91,96,97, 98,99, 100,28, 47,49,50				
A5	Isolation, unused to unused	91	96,97,98, 99,100, 28,47,49, 50				
A6	Isolation, unused to unused	96	97,98,99, 100,28, 47,49,50				
A7	Isolation, unused to unused	97	98,99, 100,28, 47,49,50				
A8	Isolation, unused to unused	98	99,100, 28,47,49, 50				
A9	Isolation, unused to unused	99	100,28, 47,49,50				
A10	Isolation, unused to unused	100	28,47,49, 50	10	INF	MOhm	1-7

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TABLE II. Isolation test - Continued.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A11	Isolation, unused to unused	28	47,49,50	10	INF	MOhm	1-7
A12	Isolation, unused to unused	47	49,50				
A13	Isolation, unused to unused	49	50	10	INF	MOhm	1-7

NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A1-A13
Isolation measurement (see note 6)	NA NA	± 500 KOhm ± 50 KOhm	A1-A13 A1-A13

2. The ohmmeter used shall be item R06.

3. In the setup conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, and 'B' shall correspond to the negative terminal.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

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NOTES: - Continued.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. The minimum end-of-life limit is 1 MOhm.

7. Abbreviations used are defined in 6.1.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, Vcc	1	31	0.0	1.0	Ohm	1-4
B2	Continuity, GND	10	15				
B3	Continuity, GND	15	25				
B4	Continuity, GND	25	40	0.0	1.0	Ohm	1-4
B5	Continuity, Vcc	1	31	0.0	0.1	Ohm	1-4
B6	Continuity, GND	10	15				
B7	Continuity, GND	15	25				
B8	Continuity, GND	25	40	0.0	0.1	Ohm	1-4

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA NA	± 50 mOhm ± 5 mOhm	B1-B4 B5-B8

2. The ohmmeter used shall be item R07.

3. In the setup conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. Abbreviations used are defined in 6.1.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect Positive lead to	Connect Negative lead to	25 ± 5°C Initial limits	Notes
C	Vcc	7.0	1,31	10,15 25,40	Module shall pass all remaining tests.	1-5

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	966 mA	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. Abbreviations used are defined in 6.1.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	D0	62	5.5	0.0	100.0	uA	1-3
D2	D1	64					
D3	D2	12					
D4	D3	4					
D5	D4	5					
D6	D5	60					
D7	D6	70					
D8	D7	74					
D9	D8	20					
D10	D9	19					
D11	D10	11					
D12	D11	73					
D13	D12	75					
D14	D13	76					
D15	D14	81					
D16	D15	87					
D17	D16	95					
D18	D17	45					
D19	-RST0	54					
D20	-RST1	56	5.5	0.0	100.0	uA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	-RST2	55	5.5	0.0	100.0	uA	1-3
D22	-RST3	51	5.5	0.0	100.0	uA	1-3
D23	24MHZIN	37	5.5	0.0	1.0	mA	1-3
D24	1A	86					
D25	2A	35	5.5	0.0	1.0	mA	1-3
D26	PADDR5	24	5.5	0.0	50.0	uA	1-3
D27	PADDR6	44					
D28	PADDR8	92					
D29	PADDR9	26					
D30	-PROM OE	41	5.5	0.0	50.0	uA	1-3
D31	D18	63	5.5	0.0	150.0	uA	1-3
D32	D19	13					
D33	D20	66					
D34	D21	16					
D35	D22	57					
D36	D23	48	5.5	0.0	150.0	uA	1-3
D37	-OE	67	5.5	0.0	300.0	uA	1-3
D38	-LATCH	84	5.5	0.0	300.0	uA	1-3
D39	CLK	78	5.5	0.0	200.0	uA	1-3
D40	-TI1	38	5.5	0.0	8.4	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	-TI2	33	5.5	0.0	7.4	mA	1-3

NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V 5.5 V 5.5 V 5.5 V	± 55 mV + 0mV,-55mV ± 55 mV + 0mV,-55mV	D1-D18 D19-D21 D22-D39 D40,D41
Power supply current limit	0.966 A	NA	D1-D41
Forcing voltage, (Vin)	5.5 V 5.5 V 5.5 V 5.5 V	± 55 mV - 0mV,+55mV ± 55mV - 0mV,+55mV	D1-D18 D19-D21 D22-D39 D40,D41
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D41 D1-D41
Input current measurement	NA NA NA	± 5.0 uA ± 50.0 uA ± 2.5 uA	D1-D22 D23-D25 D26-D30

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Input current measurement	NA	± 7.5 uA	D31-D36
	NA	± 15.0 uA	D37,D38
	NA	± 10.0 uA	D39
	NA	± 0.42 mA	D40
	NA	± 0.37 mA	D41

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	D0	62	-18.0	-1.5	-0.4	V	1-4
E2	D1	64					
E3	D2	12					
E4	D3	4					
E5	D4	5					
E6	D5	60					
E7	D6	70					
E8	D7	74					
E9	D8	20					
E10	D9	19					
E11	D10	11					
E12	D11	73					
E13	D12	75					
E14	D13	76					
E15	D14	81					
E16	D15	87					
E17	D16	95					
E18	D17	45					
E19	-RST0	54					
E20	-RST1	56	-18.0	-1.5	-0.4	V	1-4

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	-RST2	55	-18.0	-1.5	-0.4	V	1-4
E22	-RST3	51	-18.0	-1.5	-0.4	V	1-4
E23	24MHZIN	37	-18.0	-1.2	-0.4	V	1-4
E24	1A	86					
E25	2A	35	-18.0	-1.2	-0.4	V	1-4
E26	PADDR5	24	-10.0	-1.5	-0.4	V	1-4
E27	PADDR6	44					
E28	PADDR8	92					
E29	PADDR9	26					
E30	-PROM OE	41					
E31	D18	63					
E32	D19	13					
E33	D20	66					
E34	D21	16					
E35	D22	57					
E36	D23	48	-10.0	-1.5	-0.4	V	1-4
E37	-OE	67	-18.0	-1.5	-0.4	V	1-4
E38	-LATCH	84					
E39	CLK	78	-18.0	-1.5	-0.4	V	1-4
E40	-TI1	38	-18.0	-1.2	-0.4	V	1-4

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TABLE VI. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	-TI2	33	-18.0	-1.2	-0.4	V	1-4

NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 0.0 V	± 45 mV ± 100 mV	E1-E39 E40,E41
Power supply current limit	0.966 A	NA	E1-E41
Forcing current, (Iin)	-18 mA -10 mA -18 mA	± 1.8 mA ± 1.0 mA ± 1.8 mA	E1-E25 E26-E36 E37-E41
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	E1-E41 E1-E41
Input voltage measurement	NA NA NA NA	± 55 mV ± 40 mV ± 55 mV ± 40 mV	E1-E22 E23-E25 E26-E39 E40,E41

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table VI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

4. Test conditions E40 and E41 are performed with power off.

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TABLE VII. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	Q0	61	0.0	-130.0	-15.0	mA	1-4
F2	Q1	65					
F3	Q2	14					
F4	Q3	2					
F5	Q4	6					
F6	Q5	59					
F7	Q6	69					
F8	Q7	72					
F9	Q8	21					
F10	Q9	18					
F11	Q10	68					
F12	Q11	22					
F13	Q12	23					
F14	Q13	27					
F15	Q14	80					
F16	Q15	89					
F17	Q16	93					
F18	Q17	43					
F19	Q18	58					
F20	Q19	94	0.0	-130.0	-15.0	mA	1-4

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TABLE VII. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	Q20	17	0.0	-130.0	-15.0	mA	1-4
F22	Q21	9					
F23	Q22	8					
F24	Q23	3	0.0	-130.0	-15.0	mA	1-4
F25	12MHZA	30	0.0	-225.0	-50.0	mA	1-4
F26	12MHZB	32					
F27	8MHZ	82					
F28	4MHZ	29					
F29	1B	34					
F30	2B	36	0.0	-225.0	-50.0	mA	1-4
F31	PDATA0	85	0.0	-90.0	-10.0	mA	1-4
F32	PDATA1	90					
F33	PDATA2	42					
F34	PDATA3	39	0.0	-90.0	-10.0	mA	1-4
F35	6MHZ	71	0.0	-110.0	-40.0	mA	1-4
F36	-6MHZ	77	0.0	-110.0	-40.0	mA	1-4
F37	-RSTO-3	52	0.0	-130.0	-15.0	mA	1-4
F38	-RSTO-2	7	0.0	-130.0	-15.0	mA	1-4
F39	-POR	53	0.0	-225.8	-50.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F39
Power supply current limit	0.966 A	NA	F1-F39
Forcing voltage, (Vout)	0.0 V	± 20 mV	F1-F39
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F39 F1-F39
Output current measurement	NA NA NA NA NA NA	± 5.75 mA ± 8.75 mA ± 4.0 mA ± 3.5 mA ± 5.75 mA ± 8.79 mA	F1-F24 F25-F30 F31-F34 F35,F36 F37,F38 F39

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
G1	D0	62	2.7	0.0	20.0	uA	1-3
G2	D1	64					
G3	D2	12					
G4	D3	4					
G5	D4	5					
G6	D5	60					
G7	D6	70					
G8	D7	74					
G9	D8	20					
G10	D9	19					
G11	D10	11					
G12	D11	73					
G13	D12	75					
G14	D13	76					
G15	D14	81					
G16	D15	87					
G17	D16	95					
G18	D17	45	2.7	0.0	20.0	uA	1-3
G19	-RST0	54	2.7	-295.0	-247.0	uA	1-3
G20	-RST1	56	2.7	-295.0	-247.0	uA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	-RST2	55	2.7	-295.0	-247.0	uA	1-3
G22	-RST3	51	2.7	0.0	20.0	uA	1-3
G23	24MHZIN	37	2.7	0.0	50.0	uA	1-3
G24	1A	86					
G25	2A	35					
G26	PADDR5	24					
G27	PADDR6	44					
G28	PADDR8	92					
G29	PADDR9	26					
G30	-PROM OE	41	2.7	0.0	50.0	uA	1-3
G31	D18	63	2.7	0.0	70.0	uA	1-3
G32	D19	13					
G33	D20	66					
G34	D21	16					
G35	D22	57					
G36	D23	48	2.7	0.0	70.0	uA	1-3
G37	-OE	67	2.7	0.0	60.0	uA	1-3
G38	-LATCH	84	2.7	0.0	60.0	uA	1-3
G39	CLK	78	2.7	0.0	40.0	uA	1-3
G40	-TI1	38	2.7	-2.95	-1.99	mA	1-3

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TABLE VIII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	-TI2	33	2.7	-2.95	-1.89	mA	1-3

NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G41
Power supply current limit	0.966 A	NA	G1-G41
Forcing voltage, (Vih)	2.7 V	± 135 mV	G1-G41
Pattern voltage high, (VH)	3.0 V	± 100 mV	G1-G41
low, (VL)	0.0 V	± 100 mV	G1-G41
Input current measurement	NA	± 1.0 uA	G1-G18
	NA	± 2.4 uA	G19-G21
	NA	± 1.0 uA	G22
	NA	± 2.5 uA	G23-G30
	NA	± 3.5 uA	G31-G36
	NA	± 3.0 uA	G37,G38
	NA	± 2.0 uA	G39
	NA	± 48.0 uA	G40
	NA	± 53.0 uA	G41

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H1	D0	62	0.4	-400.0	0.0	uA	1-3
H2	D1	64					
H3	D2	12					
H4	D3	4					
H5	D4	5					
H6	D5	60					
H7	D6	70					
H8	D7	74					
H9	D8	20					
H10	D9	19					
H11	D10	11					
H12	D11	73					
H13	D12	75					
H14	D13	76					
H15	D14	81					
H16	D15	87					
H17	D16	95					
H18	D17	45	0.4	-400.0	0.0	uA	1-3
H19	-RST0	54	0.4	-917.0	-516.0	uA	1-3
H20	-RST1	56	0.4	-917.0	-516.0	uA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H21	-RST2	55	0.4	-917.0	-516.0	uA	1-3
H22	-RST3	51	0.4	-380.0	-30.0	uA	1-3
H23	24MHZIN	37	0.5	-400.0	0.0	uA	1-3
H24	1A	86					
H25	2A	35	0.5	-400.0	0.0	uA	1-3
H26	PADDR5	24	0.5	-250.0	-1.0	uA	1-3
H27	PADDR6	44					
H28	PADDR8	92					
H29	PADDR9	26					
H30	-PROM OE	41	0.5	-250.0	-1.0	uA	1-3
H31	D18	63	0.4	-650.0	-1.0	uA	1-3
H32	D19	13					
H33	D20	66					
H34	D21	16					
H35	D22	57					
H36	D23	48	0.4	-650.0	-1.0	uA	1-3
H37	-OE	67	0.4	-1260.0	0.0	uA	1-3
H38	-LATCH	84	0.4	-1200.0	0.0	uA	1-3
H39	CLK	78	0.4	-630.0	0.0	uA	1-3
H40	-TI1	38	0.4	-40.9	-11.0	mA	1-3

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TABLE IX. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
H41	-TI2	33	0.4	-37.3	-9.8	mA	1-3

NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	H1-H41
Power supply current limit	0.966 A	NA	H1-H41
Forcing voltage, (Vil)	0.4 V 0.5 V 0.4 V	± 20 mV ± 25 mV ± 20 mV	H1-H22 H23-H30 H31-H41
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	H1-H41 H1-H41
Input current measurement	NA NA NA NA NA NA NA NA NA NA NA	± 20.0 uA ± 20.0 uA ± 17.5 uA ± 20.0 uA ± 12.4 uA ± 32.45 uA ± 63.0 uA ± 60.0 uA ± 31.5 uA ± 1.5 mA ± 1.4 mA	H1-H18 H19-H21 H22 H23-H25 H26-H30 H31-H36 H37 H38 H39 H40 H41

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table IX and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE X. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	Q0	61	-1.0	2.4	4.5	V	1-3
J2	Q1	65					
J3	Q2	14					
J4	Q3	2					
J5	Q4	6					
J6	Q5	59					
J7	Q6	69					
J8	Q7	72					
J9	Q8	21					
J10	Q9	18					
J11	Q10	68					
J12	Q11	22					
J13	Q12	23					
J14	Q13	27					
J15	Q14	80					
J16	Q15	89					
J17	Q16	93					
J18	Q17	43					
J19	Q18	58					
J20	Q19	94	-1.0	2.4	4.5	V	1-3

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	Q20	17	-1.0	2.4	4.5	V	1-3
J22	Q21	9					
J23	Q22	8					
J24	Q23	3	-1.0	2.4	4.5	V	1-3
J25	12MHZA	30	-3.0	2.4	4.5	V	1-3
J26	12MHZB	32					
J27	8MHZ	82					
J28	4MHZ	29					
J29	1B	34					
J30	2B	36	-3.0	2.4	4.5	V	1-3
J31	PDATA0	85	-2.0	2.4	4.5	V	1-3
J32	PDATA1	90					
J33	PDATA2	42					
J34	PDATA3	39	-2.0	2.4	4.5	V	1-3
J35	6MHZ	71	-1.0	2.5	4.5	V	1-3
J36	-6MHZ	77	-1.0	2.5	4.5	V	1-3
J37	-RSTO-3	52	-0.4	2.5	4.5	V	1-3
J38	-RSTO-2	7	-0.4	2.5	4.5	V	1-3
J39	-POR	53	-2.96	2.4	4.5	V	1-3
J40	24MHZOUT	88	-0.24	2.4	4.5	V	1-3

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TABLE X. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	PU	46	-1.0	2.4	2.6	V	1-3

NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J41
Power supply current limit	0.966 A	NA	J1-J41
Forcing current, (Ioh)	-1.0 mA -3.0 mA -2.0 mA -1.0 mA -0.4 mA -2.96 mA -0.24 mA -1.0 mA	± 10.0 uA ± 30.0 uA ± 20.0 uA ± 10.0 uA ± 4.0 uA ± 29.6 uA ± 2.4 uA ± 10.0 uA	J1-J24 J25-J30 J31-J34 J35,J36 J37,J38 J39 J40 J41
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	J1-J41 J1-J41
Output voltage measurement	NA NA NA NA	± 105 mV ± 100 mV ± 105 mV ± 10 mV	J1-J34 J35-J38 J39,J40 J41

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

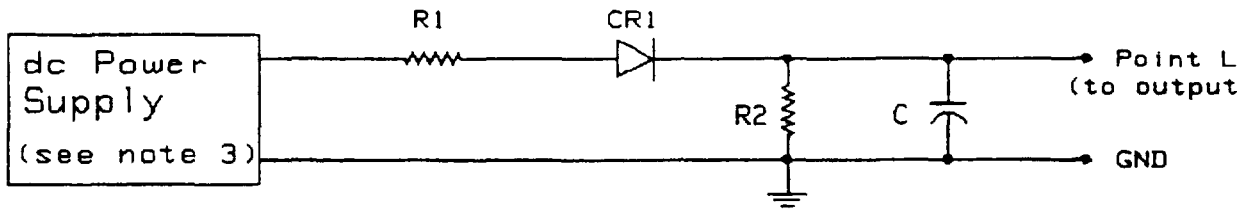
(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 150 ± 15 pF (including probe and parasitic capacitance) for sample dynamic tests and output pulse width and pulse period tests.

(b) C = 50 ± 5 pF (including probe and parasitic capacitance) for crosstalk and disable time tests.

(c) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(d) The functional test shall be performed with only a 150 ± 15 pF load (including probe and parasitic capacitance) attached to each output.

(e) C not required for any other tests.

3. The load power supply voltage shall be 2.1 V ± 1 percent for the output disable time test and 2.5 V ± 1 percent for all other tests.

FIGURE 7. Logic load circuit.

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NOTES: - Continued.

4. Select R1 for a current flow of Iol when point L is held at Vol with the load power supply set at 2.5 V ± 1 percent. The values of Iol and Vol for each output are listed below.

5. The value of R2 for each output load shall be as specified below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
61,65,14,2,6,69, 59,72,21,18,68, 22,23,27,80,89, 93,43,58,94,17, 9,8,3	2400	12.0	0.4
30,32,82,29,34,36	800	48.0	0.55
85,90,42,39	1200	12.0	0.5
71,77	2500	20.0	0.5
7,52	6250	4.0	0.4
53	811	47.2	0.55
88	10000	9.6	0.5
46	2400	NA	NA

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE XI. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
K1	Q0	61	12.0	0.0	0.4	V	1-3
K2	Q1	65					
K3	Q2	14					
K4	Q3	2					
K5	Q4	6					
K6	Q5	59					
K7	Q6	69					
K8	Q7	72					
K9	Q8	21					
K10	Q9	18					
K11	Q10	68					
K12	Q11	22					
K13	Q12	23					
K14	Q13	27					
K15	Q14	80					
K16	Q15	89					
K17	Q16	93					
K18	Q17	43					
K19	Q18	58					
K20	Q19	94	12.0	0.0	0.4	V	1-3

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TABLE XI. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
K21	Q20	17	12.0	0.0	0.4	V	1-3
K22	Q21	9					
K23	Q22	8					
K24	Q23	3	12.0	0.0	0.4	V	1-3
K25	12MHZA	30	48.0	0.0	0.55	V	1-3
K26	12MHZB	32					
K27	8MHZ	82					
K28	4MHZ	29					
K29	1B	34					
K30	2B	36	48.0	0.0	0.55	V	1-3
K31	PDATA0	85	12.0	0.0	0.5	V	1-3
K32	PDATA1	90					
K33	PDATA2	42					
K34	PDATA3	39	12.0	0.0	0.5	V	1-3
K35	6MHZ	71	20.0	0.0	0.5	V	1-3
K36	-6MHZ	77	20.0	0.0	0.5	V	1-3
K37	-RSTO-3	52	4.0	0.0	0.4	V	1-3
K38	-RSTO-2	7	4.0	0.0	0.4	V	1-3
K39	-POR	53	47.24	0.0	0.55	V	1-3
K40	24MHZOUT	88	9.6	0.0	0.5	V	1-3

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	K1-K40
Power supply current limit	0.966 A	NA	K1-K40
Forcing current, (Iol)	12.0 mA 48.0 mA 12.0 mA 20.0 mA 4.0 mA 47.2 mA 9.6 mA	± 120 uA ± 480 uA ± 120 uA ± 200 uA ± 40 uA ± 472 uA ± 96 uA	K1-K24 K25-K30 K31-K34 K35,K36 K37,K38 K39 K40
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 K1-K40 K1-K40
Output voltage measurement	NA NA NA NA NA NA	± 20.0 mV ± 27.5 mV ± 25.0 mV ± 20.0 mV ± 27.5 mV ± 25.0 mV	K1-K24 K25-K30 K31-K36 K37,K38 K39 K40

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XI and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Output leakage current, high level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	Q0	61	2.7	-20	20	uA	1-3
L2	Q1	65					
L3	Q2	14					
L4	Q3	2					
L5	Q4	6					
L6	Q5	59					
L7	Q6	69					
L8	Q7	72					
L9	Q8	21					
L10	Q9	18					
L11	Q10	68					
L12	Q11	22					
L13	Q12	23					
L14	Q13	27					
L15	Q14	80					
L16	Q15	89					
L17	Q16	93					
L18	Q17	43					
L19	Q18	58					
L20	Q19	94	2.7	-20	20	uA	1-3

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TABLE XII. Output leakage current, high level, test - Continued.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	Q20	17	2.7	-20	20	uA	1-3
L22	Q21	9					
L23	Q22	8					
L24	Q23	3	2.7	-20	20	uA	1-3
L25	PDATA0	85	5.2	-100	100	uA	1-3
L26	PDATA1	90					
L27	PDATA2	42					
L28	PDATA3	39	5.2	-100	100	uA	1-3

NOTES:

1. Output leakage current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	L1-L28
Power supply current limit	0.966 A	NA	L1-L28
Forcing voltage, (Voh)	2.7 V 5.2 V	± 135 mV ± 260 mV	L1-L24 L25-L28
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	L1-L28 L1-L28

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA NA	± 2 uA ± 10 uA	L1-L24 L25-L28

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Output leakage current, low level, test.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M1	Q0	61	0.4	-20	20	uA	1-3
M2	Q1	65					
M3	Q2	14					
M4	Q3	2					
M5	Q4	6					
M6	Q5	59					
M7	Q6	69					
M8	Q7	72					
M9	Q8	21					
M10	Q9	18					
M11	Q10	68					
M12	Q11	22					
M13	Q12	23					
M14	Q13	27					
M15	Q14	80					
M16	Q15	89					
M17	Q16	93					
M18	Q17	43					
M19	Q18	58					
M20	Q19	94	0.4	-20	20	uA	1-3

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TABLE XIII. Output leakage current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vol (V)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
M21	Q20	17	0.4	-20	20	uA	1-3
M22	Q21	9					
M23	Q22	8					
M24	Q23	3	0.4	-20	20	uA	1-3
M25	PDATA0	85	0.5	-100	100	uA	1-3
M26	PDATA1	90					
M27	PDATA2	42					
M28	PDATA3	39	0.5	-100	100	uA	1-3

NOTES:

1. Output leakage current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	M1-M28
Power supply current limit	0.966 A	NA	M1-M28
Forcing voltage, (Vol)	0.4 V 0.5 V	± 20 mV ± 25 mV	M1-M24 M25-M28
Pattern voltage high, (VH) low, (VL)	3.0 V 0.0 V	± 100 mV ± 100 mV	M1-M28 M1-M28

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA NA	± 2 uA ± 10 uA	M1-M24 M25-M28

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XIV. Power supply current test.

Test cond	Parameter	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	30	644	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	0.966 A	NA	N
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 N N
Power supply current measurement	NA	± 30.7 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate	1-3
P2	Functional test maximum power supply voltage maximum pattern rate	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	0.966 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105.0 mV ± 155.0 mV ± 27.5 mV	 P1 P2 P1,P2

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 150 pF (see figure 7). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.4\ V$

(c) The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

3. Abbreviations used are defined in 6.1.

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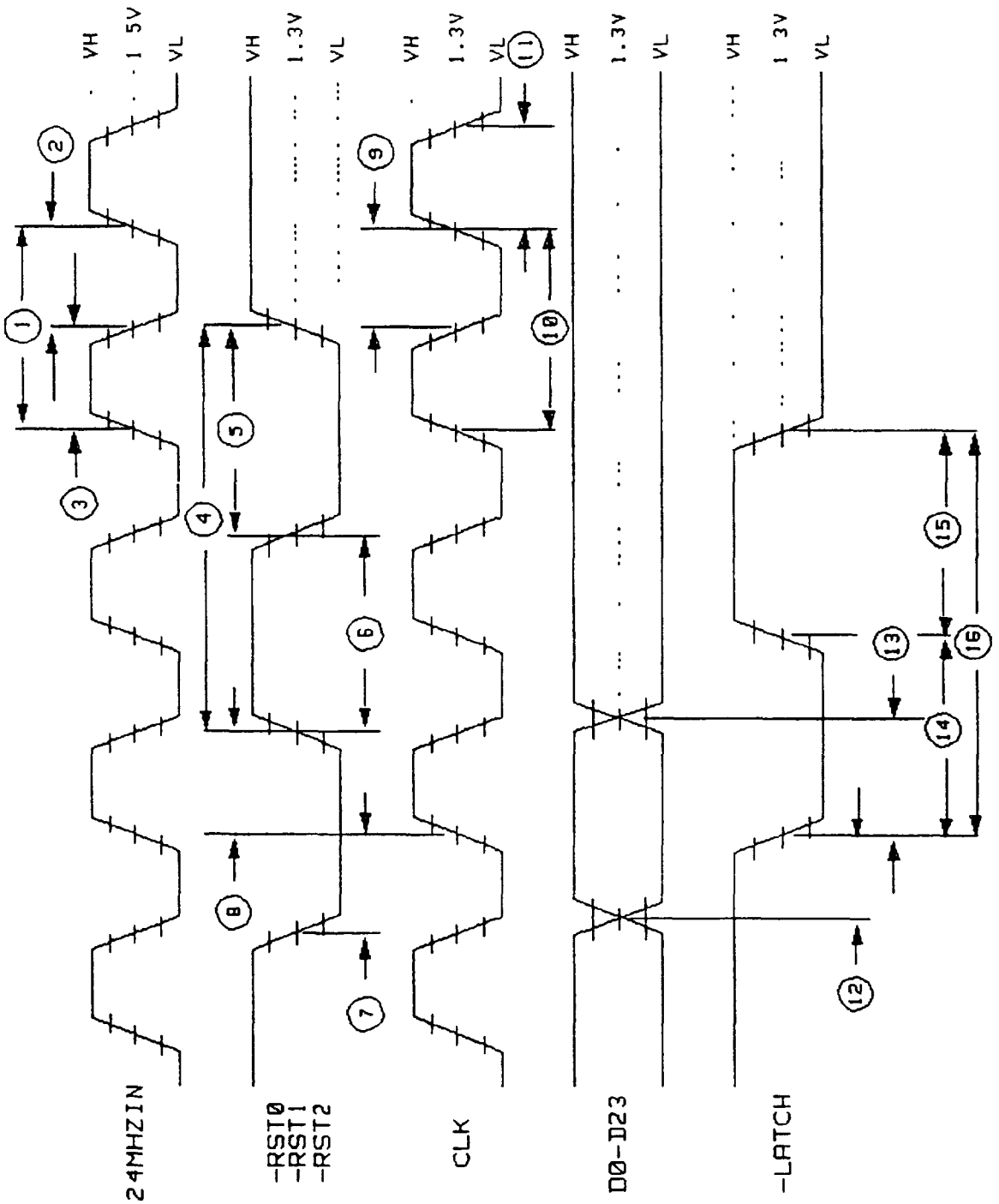


FIGURE 8. Input timing parameters for table XV.

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NOTES:

1. Input parameters:

    (a) 24MHZIN pulse period = 40 nS min.

    (b) 24MHZIN pulse width low = 20 nS min.

    (c) 24MHZIN pulse width high = 20 nS min.

    (d) -RST0, -RST1, -RST2 pulse period = 180 nS min.

    (e) -RST0, -RST1, -RST2 pulse width low = 90 nS min.

    (f) -RST0, -RST1, -RST2 pulse width high = 90 nS min.

    (g) -RST0, -RST1, -RST2 setup time = 87 nS min.

    (h) -RST0, -RST1, -RST2 hold time = 0 nS min.

    (i) CLK pulse width low = 80 nS min.

    (j) CLK pulse period = 160 nS min.

    (k) CLK pulse width high = 80 nS min.

    (l) D0-D23 setup time = 5 nS min.

    (m) D0-D23 hold time = 5 nS min.

    (n) -LATCH pulse width low = 65 nS min.

    (o) -LATCH pulse width high = 65 nS min.

    (p) -LATCH pulse period = 130 nS min.

2. Pattern voltages shall be as specified in the test tolerance table.

3. Transition times (as measured between the 0.7 V and 2.0 V points) shall not exceed 20 nS.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate minimum power supply	1-9
Q2	Dynamic test, sample maximum pattern rate maximum power supply	1-9
Q3	Dynamic test, qualification maximum pattern rate minimum power supply	1-9
Q4	Dynamic test, qualification maximum pattern rate maximum power supply	1-9

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	0.966 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 120 mV ± 20 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 105.0 mV	Q1,Q3
high, (Voh)	NA	± 155.0 mV	Q2,Q4
low, (Vol)	NA	± 27.5 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) The output compare levels shall be as follows:

High level: $Vref \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq Vref$

3. The load circuit for this test is specified on figure 7.

C = 150pF for test condition Q1 and Q2, parameters 1 through 336.
C = 300 pF for test conditons Q3 and Q4, parameters 1 through 32 and 177 through 489.

4. All of the input waveforms shall meet the requirements shown on figure 9.

5. This test shall be performed as indicated in table XVI. The dynamic parameters are found in table XVII.

6. Limits for parameters 20, 21, 22, 26, 27 and 28 are dependent on the operating frequency of clock and have been calculated using the maximum clock frequency of 6 MHz. For other clock frequencies, the minimum limit is 7/F, the maximum limit for 0 to 60°C is (8/F)+216 nS, and the maximum limit for 25°C is (8/F)+160 nS (where F = operating frequency of clock).

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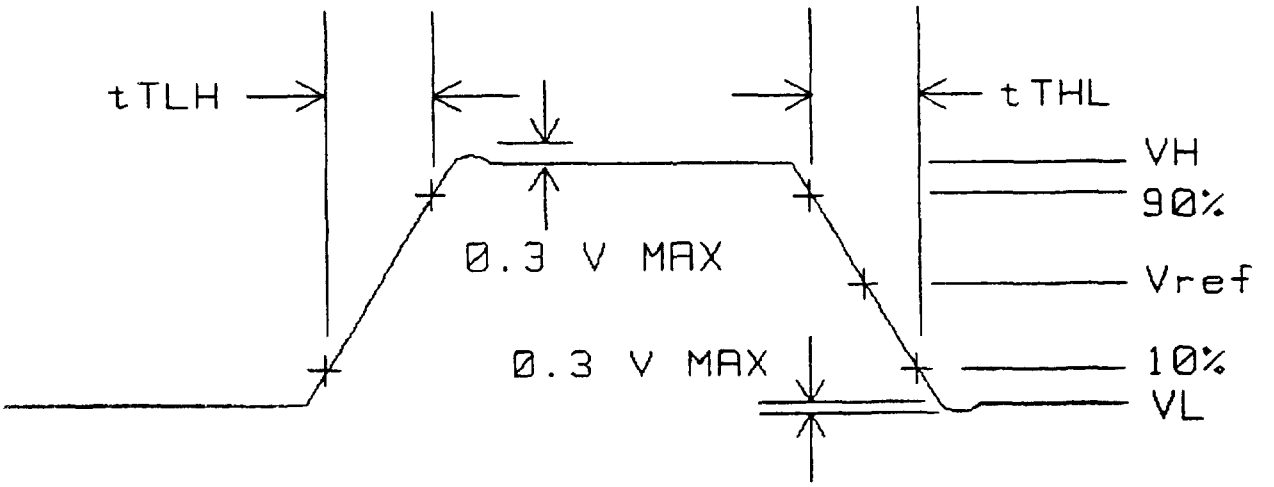
NOTES: - Continued.

7. Tests are performed with 24MHZIN connected to 24MHZOUT.

8. Input/Output (I/O) waveforms are as referenced on figure 10.

9. Abbreviations used are defined in 6.1.

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NOTES:

1. VH and VL are as specified in the test tolerance table.
2. See figures 10a through 10f for specific value of Vref.
3. Rise and fall times for the above waveform are as follows:

Input pin	tTLH	tTHL
37,86,35,24,44,92,26,41	2.5 ± 1.5 nS	2.5 ± 1.5 nS
54,55,56,78,51,60,11,19,64, 20,70,12,62,73,75,81,95,45, 76,74,4,5,63,13,66,16,57,48, 67,84	15.0 ± 1.5 nS	6.0 ± 1.5 nS

FIGURE 9. Sample input waveform for table XVI.

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TABLE XVII. Dynamic parameters.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
17	1-2	24MHZIN to 12MHZA, 12MHZB tPLH	-	-	-	31	29	31	nS	10a
16	3-4	24MHZIN to 12MHZA, 12MHZB tPHL	-	-	-	29	27	29	nS	10a
18	5-6	24MHZIN to 6MHZ, -6MHZ tPLH	-	-	-	35	31	35	nS	10a
19	7-8	24MHZIN to 6MHZ, -6MHZ tPHL	-	-	-	33	29	33	nS	10a
20	9	24MHZIN to 8MHZ tPLH	-	-	-	31	29	31	nS	10a
21	10	24MHZIN to 8MHZ tPHL	-	-	-	29	27	29	nS	10a
22	11	24MHZIN to 4MHZ tPLH	-	-	-	84	68	84	nS	10a
23	12	24MHZIN to 4MHZ tPHL	-	-	-	72	58	72	nS	10a
51	13	1A to 1B tPLH	-	-	-	13	13	13	nS	10f

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TABLE XVII. Dynamic parameters - Continued

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
50	14	1A to 1B tPHL	-	-	-	11	11	11	nS	10f
51	15	2A to 2B tPLH	-	-	-	13	13	13	nS	10f
50	16	2A to 2B tPHL	-	-	-	11	11	11	nS	10f
36	17-19	-RST0, -RST1, -RST2 to -RST0-2 tPHL	-	-	-	113	91	113	nS	10c
38	20-22	-RST0, -RST1, -RST2 to -RST0-2 tPLH	1167	1167	1167	1549	1493	1549	nS	10c
36	23-25	-RST0, -RST1, -RST2 to -RST0-3 tPHL	-	-	-	113	91	113	nS	10c
38	26-28	-RST0, -RST1, -RST2 to -RST0-3 tPLH	1167	1167	1167	1549	1493	1549	nS	10c

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
37	29-30	CLK to -RST0-2, -RST0-3 tPLH	-	-	-	109	85	109	nS	10c
53	31	-RST3 to -RST0-3 tPHL	-	-	-	41	31	41	nS	10f
52	32	-RST3 to -RST0-3 tPLH	-	-	-	37	27	37	nS	10f
44	33-56	D0-D23 to Q0-Q23 tPLH	-	-	-	36	30	36	nS	10d
44	57-80	D0-D23 to Q0-Q23 tPHL	-	-	-	35	29	35	nS	10d
45	81-104	-LATCH to Q0-Q23 tPLH	-	-	-	51	42	51	nS	10d
45	105-128	-LATCH to Q0-Q23 tPHL	-	-	-	50	41	50	nS	10d
46	129-152	-OE to Q0-Q23 tPZL	-	-	-	58	47	58	nS	10d

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
46	153-176	-OE to Q0-Q23 tPZH	-	-	-	52	43	52	nS	10d
48	177-181	D18, D20-D23 to PDATA0 tPLH	-	-	-	91	91	91	nS	10e
48	182-186	D18, D20-D23 to PDATA0 tPHL	-	-	-	89	89	89	nS	10e
48	187-192	D18-D23 to PDATA1 tPLH	-	-	-	91	91	91	nS	10e
48	193-198	D18-D23 to PDATA1 tPHL	-	-	-	89	89	89	nS	10e
48	199-201	D21-D23 to PDATA2 tPLH	-	-	-	91	91	91	nS	10e
48	202-204	D21-D23 to PDATA2 tPHL	-	-	-	89	89	89	nS	10e
48	205-208	D20-D23 to PDATA3 tPLH	-	-	-	91	91	91	nS	10e

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TABLE XVII. Dynamic parameters - Continued

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
48	209-212	D20-D23 to PDATA3 tPHL	-	-	-	89	89	89	nS	10e
47	213-220	PADDR5, PADDR6 to PDATA0-PDATA3 tPLH	-	-	-	91	91	91	nS	10e
47	221-228	PADDR5, PADDR6 to PDATA0-PDATA3 tPHL	-	-	-	89	89	89	nS	10e
47, 48	229-236	PADDR8, PADDR9 to PDATA0-PDATA3 tPLH	-	-	-	91	91	91	nS	10e
47, 48	237-244	PADDR8, PADDR9 to PDATA0-PDATA3 tPHL	-	-	-	89	89	89	nS	10e

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
49	245-248	-PROM OE to PDATA0-PDATA3 tPZH	-	-	-	46	46	46	nS	10e
49	249-252	-PROM OE to PDATA0-PDATA3 tPZL	-	-	-	44	44	44	nS	10e
40	253-276	D0-D23 to -LATCH tSETUP	5	5	5	-	-	-	nS	10d
39	277-300	D0-D23 to -LATCH tHOLD	20	20	20	-	-	-	nS	10d
25	302	-POR PWL	100	100	100	1258	1258	1258	mS	10b
29	303	-RSTO-3 PWL	100	100	100	1258	1258	1258	mS	10b
29	304	-RSTO-2 PWL	100	100	100	1258	1258	1258	mS	10b

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
24	307	Vcc at 4.5V to -POR tPLH	100	100	100	1258	1258	1258	mS	10b
31	308,309	Vcc at 4.5V to -RST0-2, -RST0-3 tPLH	100	100	100	1258	1258	1258	mS	10b
35	310-312	-RST0, -RST1, -RST2 PWL	90	90	90	-	-	-	nS	10c
43	313-315	-RST0, -RST1, -RST2 PWH	90	90	90	-	-	-	nS	10c
32	316	CLK PWL	60	60	60	-	-	-	nS	10c
33	317	CLK PWH	60	60	60	-	-	-	nS	10c
34	318-320	-RST0, -RST1, -RST2 to CLK tSETUP	61	54	61	-	-	-	nS	10c
1	324	CLK fMAX	6.0	6.0	6.0	-	-	-	MHZ	10c
2	325	-LATCH fMAX	6.67	6.67	6.67	-	-	-	MHZ	10d
42	326	-LATCH PWH	65	65	65	-	-	-	nS	10d

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TABLE XVII. Dynamic parameters - Continued.

Ref num	Parameter number	Parameter	Minimum limit			Maximum limit			Units	Fig
			0°C	25°C	60°C	0°C	25°C	60°C		
45	346-369	-LATCH to Q0-Q23 tPLH	-	-	-	69	60	69	nS	10d
45	370-393	-LATCH to Q0-Q23 tPHL	-	-	-	66	57	66	nS	10d
44	394-417	D0-D23 to Q0-Q23 tPLH	-	-	-	54	48	54	nS	10d
44	418-441	D0-D23 to Q0-Q23 tPHL	-	-	-	51	45	51	nS	10d
46	442-465	-OE to Q0-Q23 tPZL	-	-	-	74	63	74	nS	10d
46	466-489	-OE to Q0-Q23 tPZH	-	-	-	67	58	67	nS	10d

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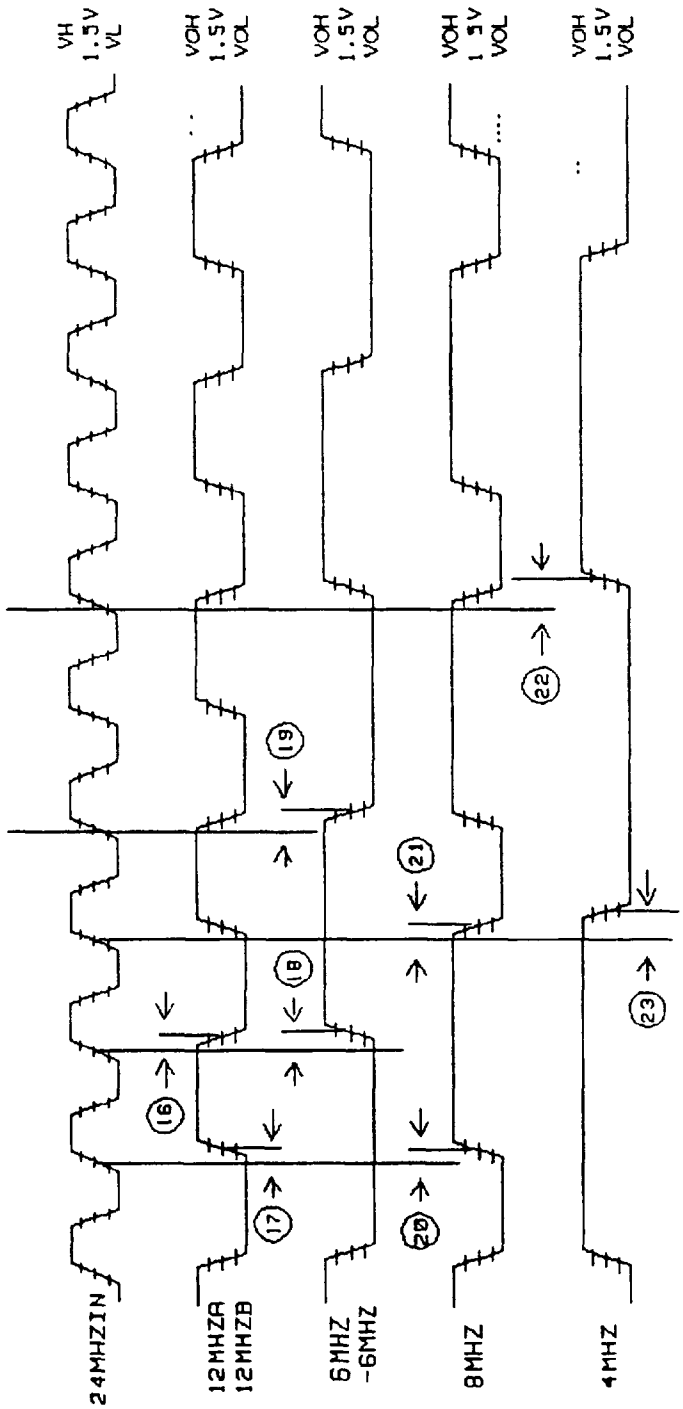


FIGURE 10a. Dynamic waveforms for table XVII.

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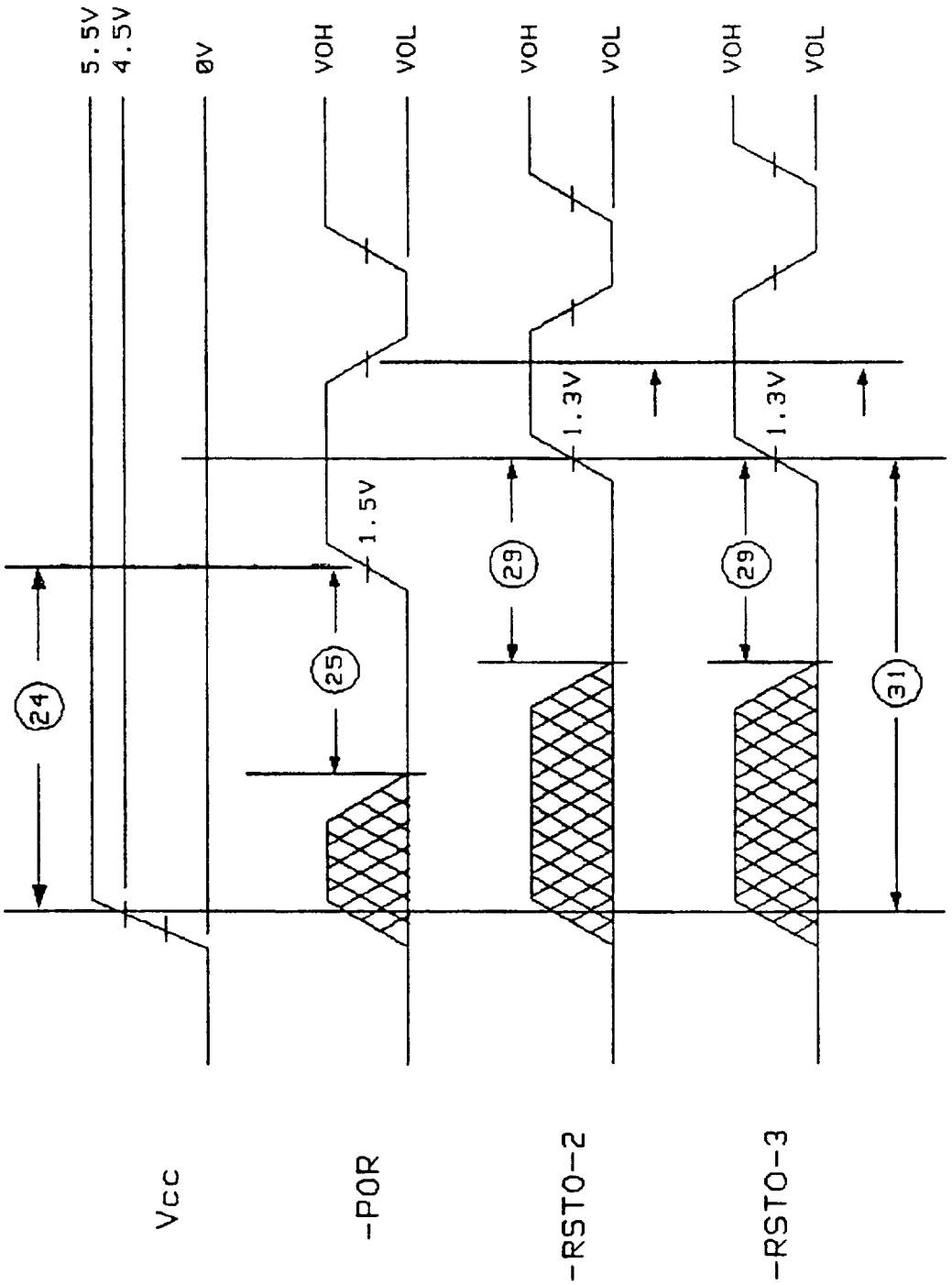


FIGURE 10b. Dynamic waveforms for table XVII.

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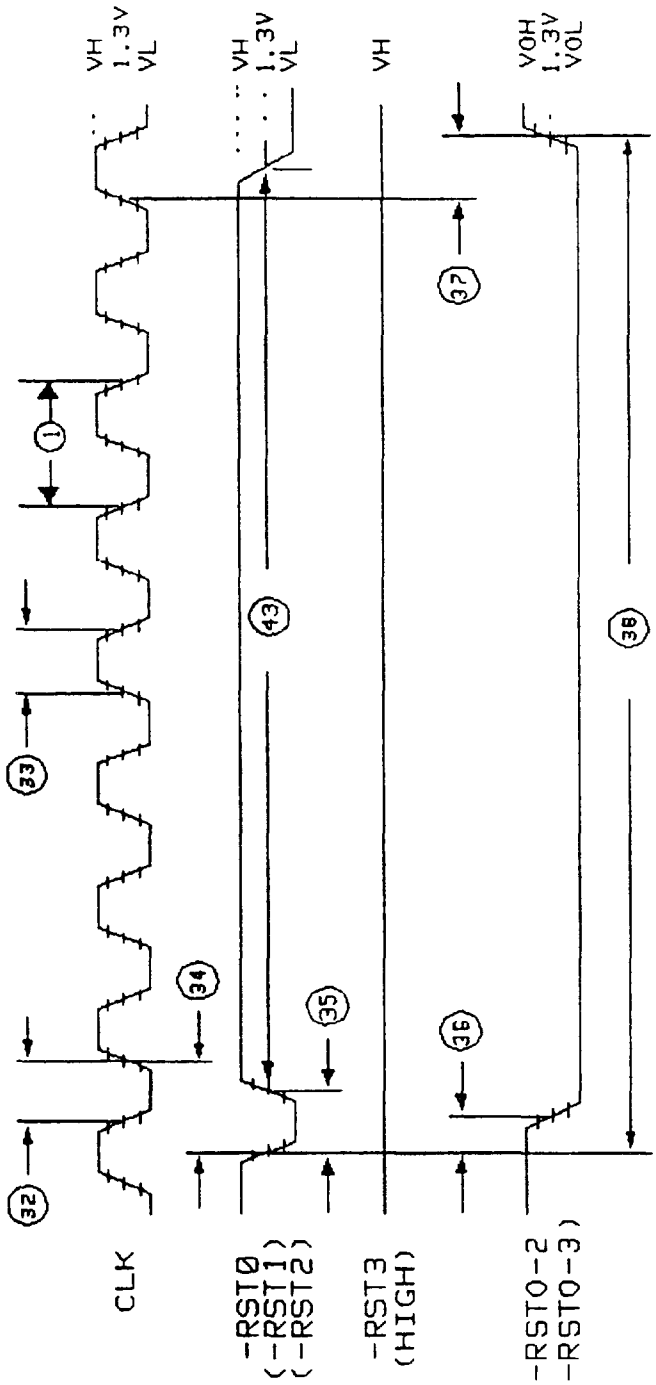


FIGURE 10c. Dynamic waveforms for table XVII.

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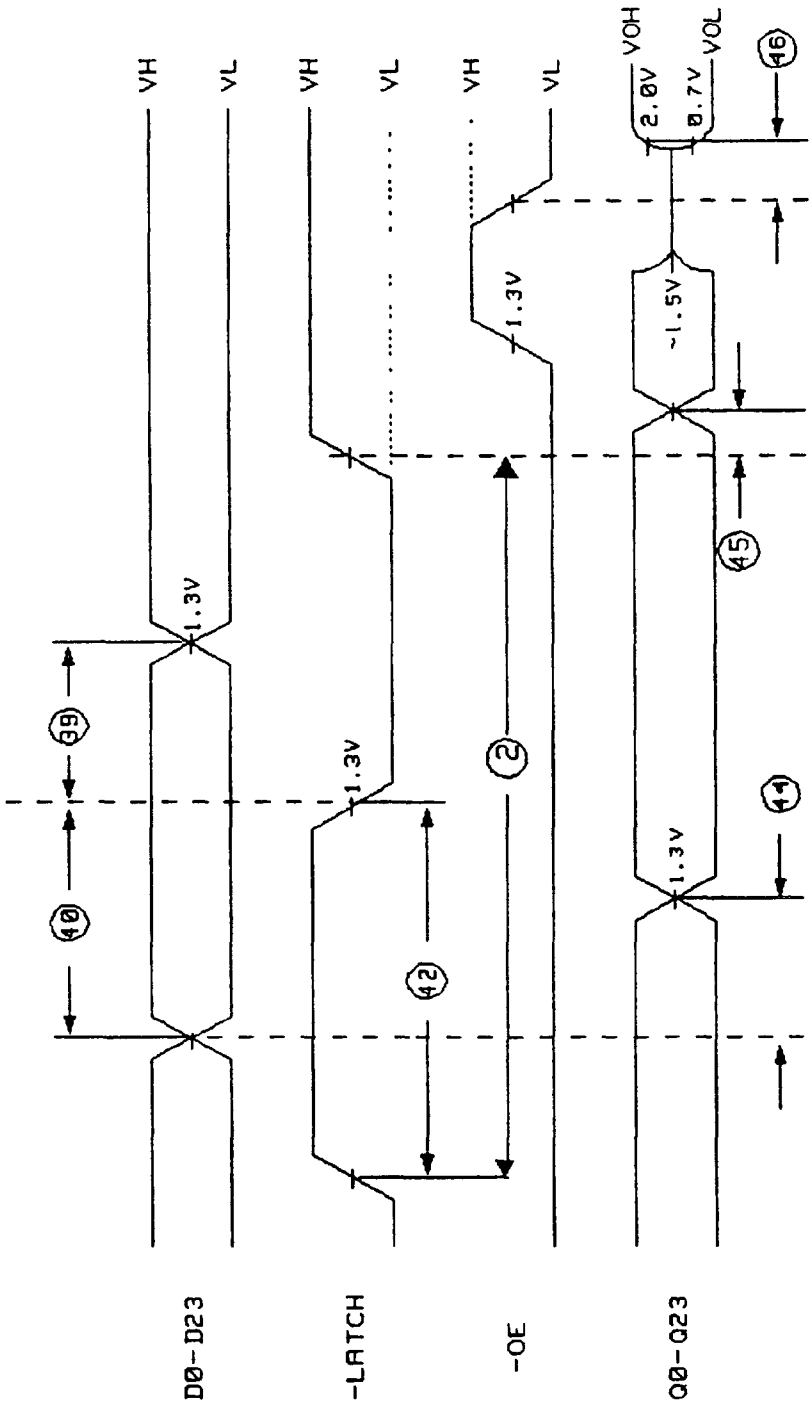


FIGURE 10d. Dynamic waveforms for table XVII.

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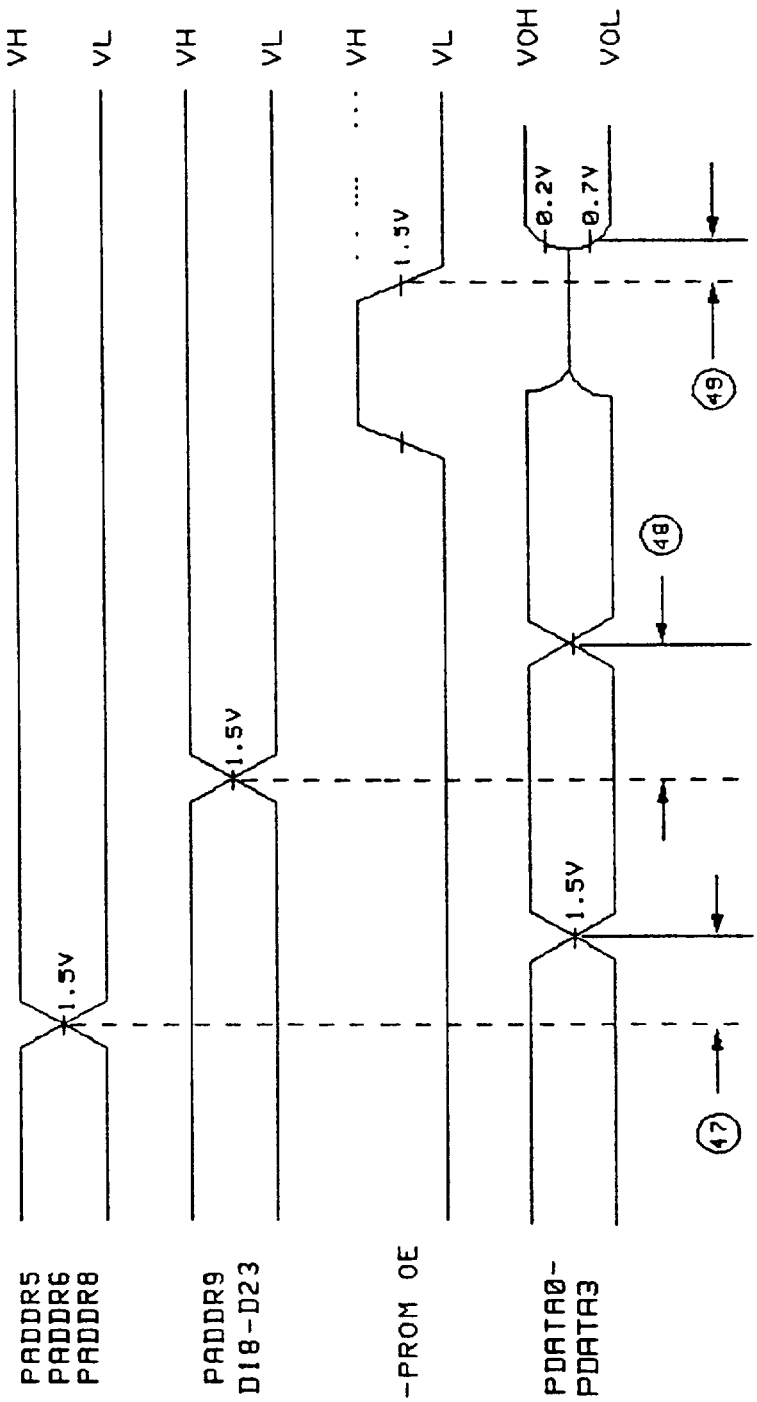


FIGURE 10e. Dynamic waveforms for table XVII.

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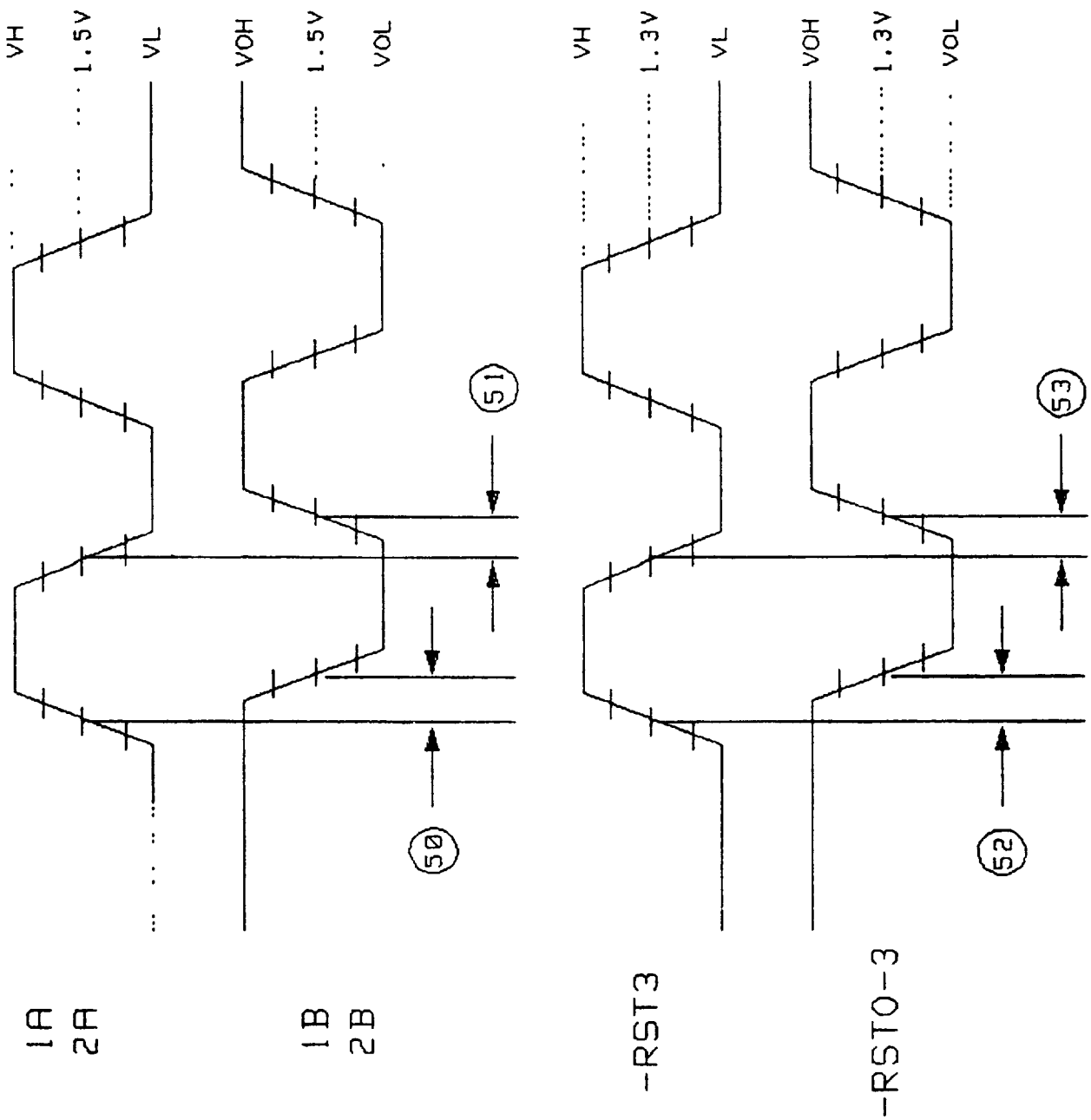


FIGURE 10f. Dynamic waveforms for table XVII.

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TABLE XVIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
R	PSFC	1,31	3.9	7.3	uF	1-3

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	R
Capacitance bridge voltage	0.3 Vp	± 30 mVp	R
Capacitance measurement	NA	± 170 nF	R

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. Abbreviations used are defined in 6.1.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
S	PSTA	1	0	500	mV	1-3

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	S
Power supply current limit	0.966 A	NA	S
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 S S
Power supply transient amplitude measurement	NA	± 20 mV	S

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX. Module inputs shall not violate the minimum input timing parameters referenced on figure 8.

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NOTES: - Continued.

(c) All Vcc pins shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 11 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. Abbreviations used are defined in 6.1.

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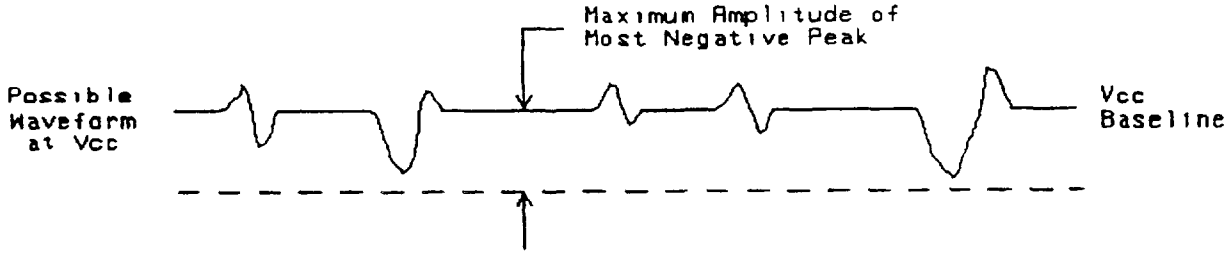


FIGURE 11. Power supply transient amplitude waveform for table XIX.

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TABLE XX. Output crosstalk, high level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
T1	Q0	61	0	800	mV	1-5	12,13
T2	Q1	65					
T3	Q2	14					
T4	Q3	2					
T5	Q4	6					
T6	Q5	59					
T7	Q6	69					
T8	Q7	72					
T9	Q8	21					
T10	Q9	18					
T11	Q10	68					
T12	Q11	22					
T13	Q12	23					
T14	Q13	27					
T15	Q14	80					
T16	Q15	89					
T17	Q16	93					
T18	Q17	43					
T19	Q18	58					
T20	Q19	94	0	800	mV	1-5	12,13

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TABLE XX. Output crosstalk, high level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Figs
T21	Q20	17	0	800	mV	1-5	12,13
T22	Q21	9					
T23	Q22	8					
T24	Q23	3	0	800	mV	1-5	12,13
T25	12MHZA	30	0	500	mV	1-5	12,13
T26	12MHZB	32					
T27	8MHZ	82					
T28	4MHZ	29					
T29	6MHZ	71					
T30	-6MHZ	77					
T31	PDATA0	85					
T32	PDATA1	90					
T33	PDATA2	42					
T34	PDATA3	39	0	500	mV	1-5	12,13
T35	-RSTO-3	52	0	400	mV	1-5	12,13
T36	-RSTO-2	7	0	400	mV	1-5	12,13
T37	-POR	53	0	500	mV	1-5	12,13
T38	PU	46	0	400	mV	1-5	12,13
T39	1B	34	0	500	mV	1-5	12,13
T40	2B	36	0	500	mV	1-5	12,13

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NOTES:

1. Output crosstalk, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	T1-T40
Power supply current limit	0.966 A	NA	T1-T40
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1-T40 T1-T40
Output crosstalk measurement	NA NA NA NA NA NA	± 40 mV ± 25 mV ± 20 mV ± 25 mV ± 20 mV ± 25 mV	T1-T24 T25-T34 T35,T36 T37 T38 T39,T40

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

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NOTES: - Continued.

(d) For each output under test in table XX, connect probe 'B' and visually inspect the output for the most negative spike as indicated on figure 12. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 7.

4. The test circuit shall be as indicated on figure 13.

5. Abbreviations used are defined in 6.1.

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TABLE XXI. Output crosstalk, low level, test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U1	Q0	61	0	800	mV	1-5	12,13
U2	Q1	65					
U3	Q2	14					
U4	Q3	2					
U5	Q4	6					
U6	Q5	59					
U7	Q6	69					
U8	Q7	72					
U9	Q8	21					
U10	Q9	18					
U11	Q10	68					
U12	Q11	22					
U13	Q12	23					
U14	Q13	27					
U15	Q14	80					
U16	Q15	89					
U17	Q16	93					
U18	Q17	43					
U19	Q18	58					
U20	Q19	94	0	800	mV	1-5	12,13

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TABLE XXI. Output crosstalk, low level, test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes	Figs
U21	Q20	17	0	800	mV	1-5	12,13
U22	Q21	9					
U23	Q22	8					
U24	Q23	3	0	800	mV	1-5	12,13
U25	12MHZA	30	0	500	mV	1-5	12,13
U26	12MHZB	32					
U27	8MHZ	82					
U28	4MHZ	29					
U29	6MHZ	71					
U30	-6MHZ	77					
U31	PDATA0	85					
U32	PDATA1	90					
U33	PDATA2	42					
U34	PDATA3	39	0	500	mV	1-5	12,13
U35	-RSTO-3	52	0	400	mV	1-5	12,13
U36	-RSTO-2	7	0	400	mV	1-5	12,13
U37	1B	34	0	500	mV	1-5	12,13
U38	2B	36	0	500	mV	1-5	12,13

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NOTES:

1. Output crosstalk, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	U1-U38
Power supply current limit	0.966 A	NA	U1-U38
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 U1-U38 U1-U38
Output crosstalk measurement	 NA NA NA NA	 ± 40 mV ± 25 mV ± 20 mV ± 25 mV	 U1-U24 U25-U34 U35,U36 U37,U38

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

(c) Use probe 'A' to verify the pattern voltage and pattern repetition rate of 1000 nS ± 200 nS, 50 percent duty cycle.

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NOTES: - Continued.

(d) For each output under test in table XXI, connect probe 'B' and visually inspect the output for the most positive spike as indicated on figure 12. The module inputs shall not violate the minimum input timing parameters on figure 8.

(e) The maximum probe capacitance is 10 pF.

3. All outputs shall be loaded as specified on figure 7.

4. The test circuit shall be as indicated on figure 13.

5. Abbreviations used are defined in 6.1.

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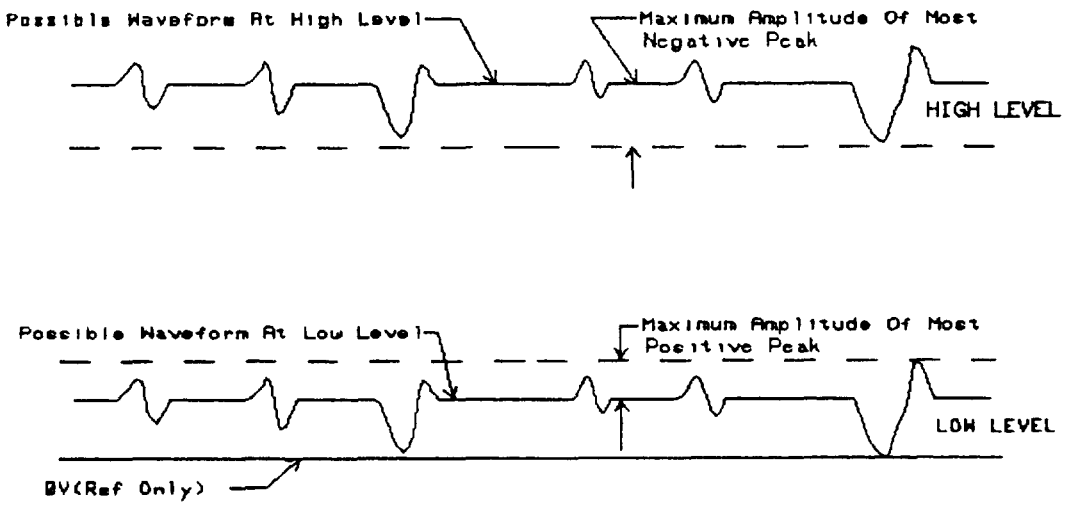


FIGURE 12. Output crosstalk, high and low level, test waveforms for tables XX and XXI.

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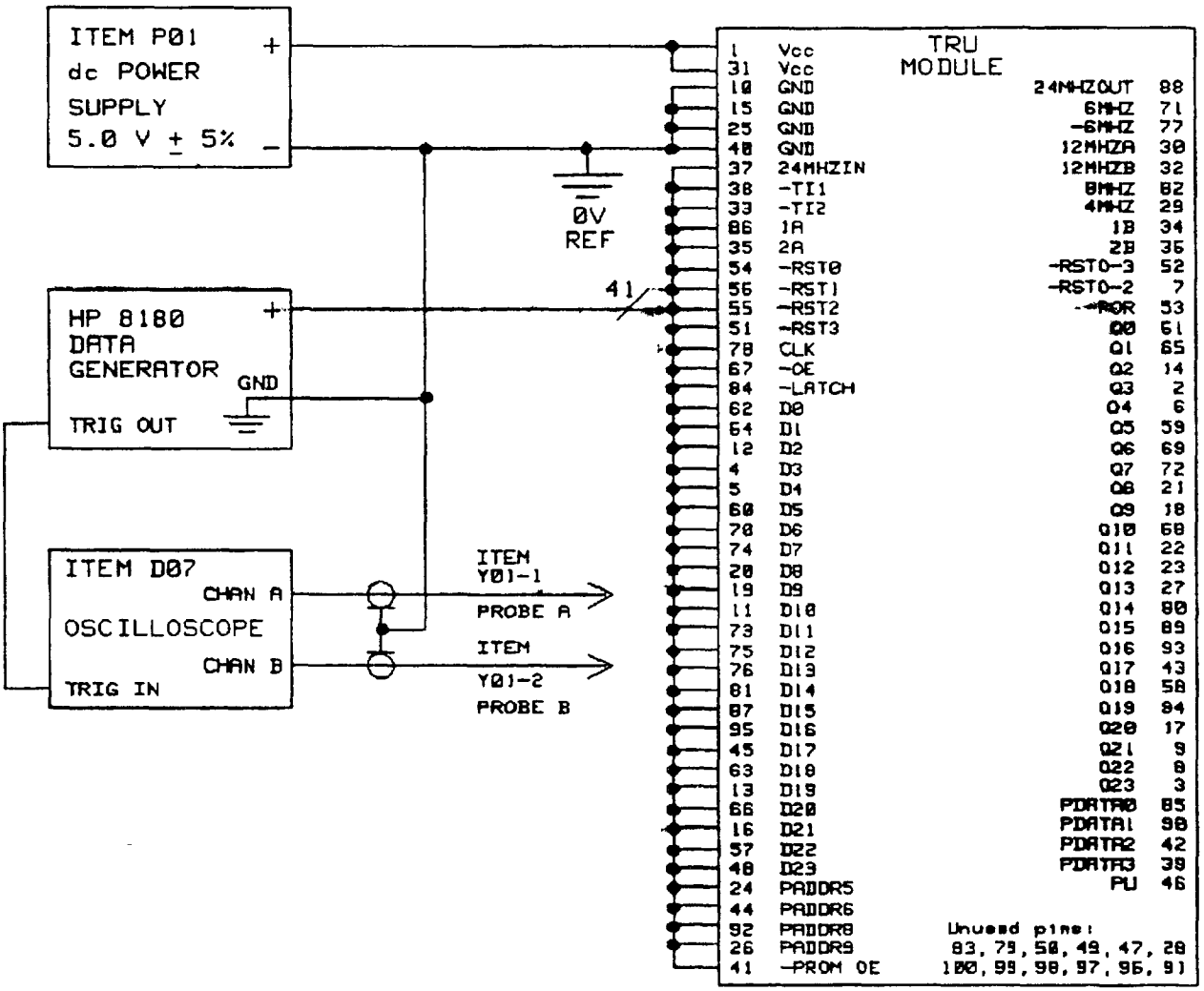


FIGURE 13. Output crosstalk, high and low level, test circuit for tables XX and XXI.

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TABLE XXII. Output crosstalk duration test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
Low level							
V1	Q0	61	0	55	nS	1-4	14
V2	Q1	65					
V3	Q2	14					
V4	Q3	2					
V5	Q4	6					
V6	Q5	59					
V7	Q6	69					
V8	Q7	72					
V9	Q8	21					
V10	Q9	18					
V11	Q10	68					
V12	Q11	22					
V13	Q12	23					
V14	Q13	27					
V15	Q14	80					
V16	Q15	89					
V17	Q16	93					
V18	Q17	43					
V19	Q18	58					
V20	Q19	94	0	55	nS	1-4	14

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TABLE XXII. Output crosstalk duration test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
V21	Q20	17	0	55	nS	1-4	14
V22	Q21	9					
V23	Q22	8					
V24	Q23	3	0	55	nS	1-4	14
High level							
V25	Q0	61	0	55	nS	1-4	14
V26	Q1	65					
V27	Q2	14					
V28	Q3	2					
V29	Q4	6					
V30	Q5	59					
V31	Q6	69					
V32	Q7	72					
V33	Q8	21					
V34	Q9	18					
V35	Q10	68					
V36	Q11	22					
V37	Q12	23					
V38	Q13	27					
V39	Q14	80					
V40	Q15	89	0	55	nS	1-4	14

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TABLE XXII. Output crosstalk duration test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig
V41	Q16	93	0	55	nS	1-4	14
V42	Q17	43					
V43	Q18	58					
V44	Q19	94					
V45	Q20	17					
V46	Q21	9					
V47	Q22	8					
V48	Q23	3	0	55	nS	1-4	14

NOTES:

1. Output crosstalk duration test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 0.25 V	V1-V48
Power supply current limit	0.966 A	NA	V1-V48
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 V1-V48 V1-V48
Crosstalk duration measurement	NA	± 2.75 nS	V1-V48

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

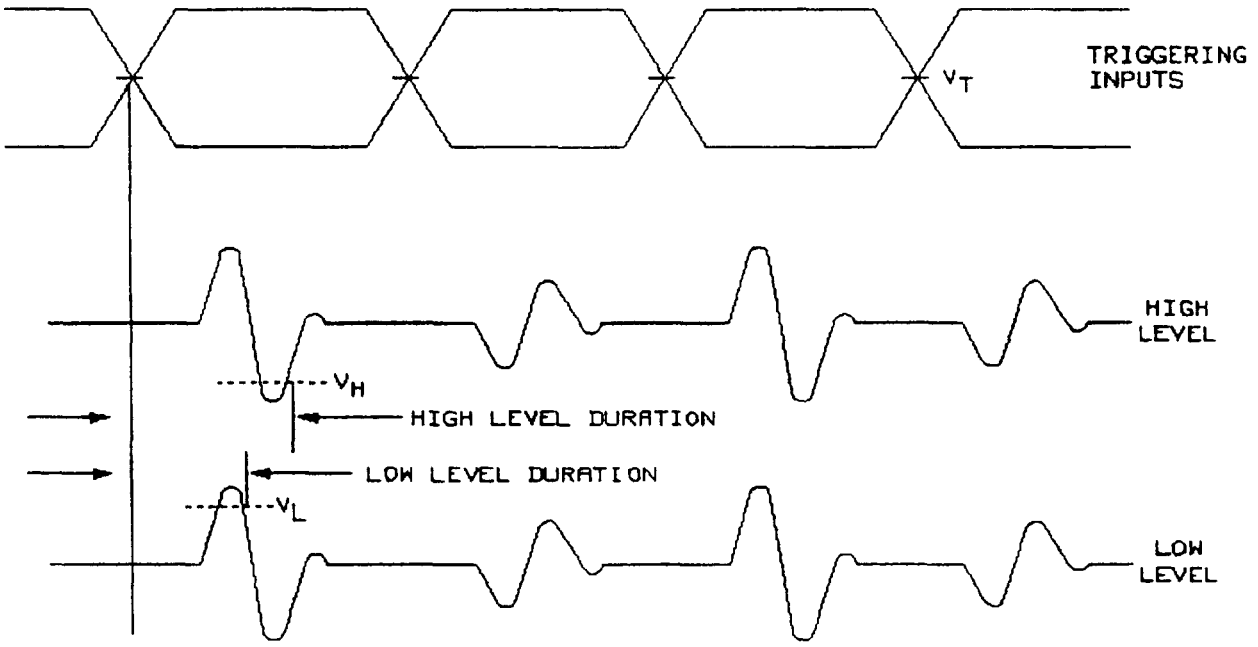
(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

(c) The duration shall be measured from the triggering edge of the input as shown on figure 14.

3. All outputs shall be loaded as specified on figure 7.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. VT = 1.5 V for test conditions V1-V24 and 1.3 V for test conditions V25-V48. VL = 400 mV (ac coupled) for test conditions V1-V24. VH = -400 mV (ac coupled) for test conditions V25-V48.

2. The duration is measured from the triggering edge of the input to the point where the output crosstalk returns to the specified voltage. If the crosstalk amplitude on a given output is less than VL or VH, the crosstalk duration should be recorded as "0".

FIGURE 14. Output crosstalk duration test waveform for table XXII.

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TABLE XXIII. Input capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
W1	D3	4	2	40	pF	1-3
W2	D4	5				
W3	D10	11				
W4	D2	12				
W5	D9	19				
W6	D8	20	2	40	pF	1-3
W7	2A	35	2	25	pF	1-3
W8	24MHZIN	37	2	25	pF	1-3
W9	D17	45	2	40	pF	1-3
W10	-RST3	51	2	25	pF	1-3
W11	-RST0	54				
W12	-RST2	55				
W13	-RST1	56	2	25	pF	1-3
W14	D5	60	2	40	pF	1-3
W15	D0	62				
W16	D1	64				
W17	D6	70				
W18	D11	73				
W19	D7	74				
W20	D12	75	2	40	pF	1-3

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TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W21	D13	76	2	40	pF	1-3
W22	D14	81	2	40	pF	1-3
W23	1A	86	2	25	pF	1-3
W24	D15	87	2	40	pF	1-3
W25	D16	95				
W26	PADDR5	24				
W27	PADDR9	26	2	40	pF	1-3
W28	-PROM OE	41	2	30	pF	1-3
W29	PADDR6	44	2	40	pF	1-3
W30	PADDR8	92	2	40	pF	1-3
W31	D19	13	4	60	pF	1-3
W32	D21	16				
W33	D22	57				
W34	D18	63				
W35	D20	66				
W36	D23	48	4	60	pF	1-3
W37	-OE	67	6	60	pF	1-3
W38	-LATCH	84	6	60	pF	1-3
W39	CLK	78	2	20	pF	1-3
W40	-TI1	38	20	150	pF	1-3

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TABLE XXIII. Input capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
W41	-TI2	33	30	160	pF	1-3

NOTES:

1. Input capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	W1-W41
Capacitance meter voltage	15 mV	± 1.5 mV	W1-W41
Capacitance measurement	NA	± 1.9 pF	W1-W6
	NA	± 1.15 pF	W7,W8
	NA	± 1.9 pF	W9
	NA	± 1.15 pF	W10-W13
	NA	± 1.9 pF	W14-W22
	NA	± 1.15 pF	W23
	NA	± 1.9 pF	W24-W27
	NA	± 1.4 pF	W28
	NA	± 1.9 pF	W29,W30
	NA	± 2.8 pF	W31-W36
	NA	± 2.7 pF	W37,W38
	NA	± 0.9 pF	W39
	NA	± 6.5 pF	W40
	NA	± 6.5 pF	W41

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NOTES: - Continued.

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIII.

3. Abbreviations used are defined in 6.1.

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TABLE XXIV. Output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
X1	Q0	61	2	50	pF	1-3
X2	Q1	65				
X3	Q2	14				
X4	Q3	2				
X5	Q4	6				
X6	Q5	59				
X7	Q6	69				
X8	Q7	72				
X9	Q8	21				
X10	Q9	18				
X11	Q10	68				
X12	Q11	22				
X13	Q12	23				
X14	Q13	27				
X15	Q14	80				
X16	Q15	89				
X17	Q16	93				
X18	Q17	43				
X19	Q18	58				
X20	Q19	94	2	50	pF	1-3

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TABLE XXIV. Output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
X21	Q20	17	2	50	pF	1-3
X22	Q21	9				
X23	Q22	8				
X24	Q23	3	2	50	pF	1-3
X25	PDATA0	85	2	35	pF	1-3
X26	PDATA1	90				
X27	PDATA2	42				
X28	PDATA3	39	2	35	pF	1-3

NOTES:

1. Output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	X1-X28
Capacitance meter voltage	15 mV	± 1.5 mV	X1-X28
Capacitance measurement	NA NA	± 2.4 pF ± 1.65 pF	X1-X24 X25-X28

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XXIV.

3. Abbreviations used are defined in 6.1.

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TABLE XXV. Threshold test.

Test cond	Parameter	Notes	Fig
Y	Threshold	1-4	8

NOTES:

1. Threshold test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	Y
Power supply current limit	0.966 A	NA	Y
Pattern voltage high, (VH) low, (VL)	 See note 3 See note 3		 Y Y
Output voltage measurement high, (Voh) low, (Vol)	 NA NA	 ± 105.0 mV ± 27.5 mV	 Y Y

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and verify the output patterns by monitoring the outputs and comparing the results to the output low and high levels specified in the output voltage tests. The module inputs shall not violate the minimum input timing parameters referenced on figure 8.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

3. The input threshold levels are as follows:

Input pin	(VH)	(VL)
62,64,12,4,5,60,70,74,20,19,11, 73,75,76,81,87,95,45,63,13,66, 16,57,48,54,56,55,51,67,84,78	2.0V ± 20 mV	0.7V ± 7 mV
24,44,92,26,37,86,35,41	2.0V ± 20 mV	0.8V ± 8 mV

4. Abbreviations used are defined in 6.1.

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TABLE XXVI. Disable time test.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 + 5°C Initial limits Min	25 + 5°C Initial limits Max	Units	Figs
Low to high impedance (tPLZ)								
Z1	-OE to Q0	Open	4,5,11, 12,13, 16,19, 20,45, 48,57, 60,62, 63,64, 66,70, 73,74, 75,76, 81,87,95	61	8	39	nS	15,16
Z2	-OE to Q1			65				
Z3	-OE to Q2			14				
Z4	-OE to Q3			2				
Z5	-OE to Q4			6				
Z6	-OE to Q5			59				
Z7	-OE to Q6	Open	4,5,11, 12,13, 16,19, 20,45, 48,57, 60,62, 63,64, 66,70, 73,74, 75,76, 81,87,95	69	8	39	nS	15,16

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Z8	-OE to Q7	Open	4,5,11, 12,13, 16,19, 20,45, 48,57, 60,62, 63,64, 66,70, 73,74, 75,76, 81,87,95	72	8	39	nS	15, 16
Z9	-OE to Q8			21				
Z10	-OE to Q9			18				
Z11	-OE to Q10			68				
Z12	-OE to Q11			22				
Z13	-OE to Q12			23				
Z14	-OE to Q13			27				
Z15	-OE to Q14	Open	4,5,11, 12,13, 16,19, 20,45, 48,57, 60,62, 63,64, 66,70, 73,74, 75,76, 81,87,95	80	8	39	nS	15, 16

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Z16	-OE to Q15	Open	4,5,11,12,13,16,19,20,45,48,57,60,62,63,64,66,70,73,74,75,76,81,87,95	89	8	39	nS	15,16
Z17	-OE to Q16			93				
Z18	-OE to Q17			43				
Z19	-OE to Q18			58				
Z20	-OE to Q19			94				
Z21	-OE to Q20			17				
Z22	-OE to Q21			9				
Z23	-OE to Q22			8				
Z24	-OE to Q23	Open	4,5,11,12,13,16,19,20,45,48,57,60,62,63,64,66,70,73,74,75,76,81,87,95	3	8	39	nS	15,16

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
High to high impedance (tPHZ)								
Z25	-OE to Q0	4,5,11, 12,13, 16,19, 20,45, 48,57, 60,62, 63,64, 66,70, 73,74, 75,76, 81,87,95	Open	61	8	73	nS	15,16
Z26	-OE to Q1	"	"	65	"	"	"	"
Z27	-OE to Q2	"	"	14	"	"	"	"
Z28	-OE to Q3	"	"	2	"	"	"	"
Z29	-OE to Q4	"	"	6	"	"	"	"
Z30	-OE to Q5	4,5,11, 12,13, 16,19, 20,45, 48,57, 60,62, 63,64, 66,70, 73,74, 75,76, 81,87,95	Open	59	8	73	nS	15,16

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Z31	-OE to Q6	4,5,11,12,13,16,19,20,45,48,57,60,62,63,64,66,70,73,74,75,76,81,87,95	Open	69	8	73	nS	15,16
Z32	-OE to Q7			72				
Z33	-OE to Q8			21				
Z34	-OE to Q9			18				
Z35	-OE to Q10			68				
Z36	-OE to Q11			22				
Z37	-OE to Q12			23				
Z38	-OE to Q13	4,5,11,12,13,16,19,20,45,48,57,60,62,63,64,66,70,73,74,75,76,81,87,95	Open	27	8	73	nS	15,16

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TABLE XXVI. Disable time test - Continued

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Z39	-OE to Q14	4,5,11, 12,13, 16,19, 20,45, 48,57, 60,62, 63,64, 66,70, 73,74, 75,76, 81,87,95	Open	80	8	73	nS	15,16
Z40	-OE to Q15	"	"	89	"	"	"	"
Z41	-OE to Q16	"	"	93	"	"	"	"
Z42	-OE to Q17	"	"	43	"	"	"	"
Z43	-OE to Q18	"	"	58	"	"	"	"
Z44	-OE to Q19	"	"	94	"	"	"	"
Z45	-OE to Q20	4,5,11, 12,13, 16,19, 20,45, 48,57, 60,62, 63,64, 66,70, 73,74, 75,76, 81,87,95	Open	17	8	73	nS	15,16

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Z46	-OE to Q21	4,5,11, 12,13, 16,19, 20,45, 48,57, 60,62, 63,64, 66,70, 73,74, 75,76, 81,87,95	Open	9	8	73	nS	15, 16
Z47	-OE to Q22			8				
Z48	-OE to Q23	4,5,11, 12,13, 16,19, 20,45, 48,57, 60,62, 63,64, 66,70, 73,74, 75,76, 81,87,95	Open	3	8	73	nS	15, 16
Low to high impedance (tPLZ)								
Z49	-PROM OE to PDATA0	24,26, 48,57,92	13,16, 44,63,66	85	2	40	nS	17, 18
Z50	-PROM OE to PDATA1			90				
Z51	-PROM OE to PDATA2	24,26, 48,57,92	13,16, 44,63,66	42	2	40	nS	17, 18

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TABLE XXVI. Disable time test - Continued.

Test cond	Signal name	Connect A to pin	Connect B to pin	Connect C to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Figs
Z52	-PROM OE to PDATA3	24,26, 48,57,92	13,16, 44,63,66	39	2	40	nS	17,18
High to high impedance (tPHZ)								
Z53	-PROM OE to PDATA0	13,16, 24,26, 48,63, 66,92	44,57	85	2	51	nS	17,18
Z54	-PROM OE to PDATA1			90				
Z55	-PROM OE to PDATA2	13,16, 24,26, 48,63, 66,92	44,57	42				
Z56	-PROM OE to PDATA3	16,26, 48,57, 66,92	13,24, 44,63	39	2	51	nS	17,18

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NOTES:

1. Disable time test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply	5.0 V	± 50 mV	Z1-Z56
Power supply current limits	0.966 A	NA	Z1-Z56
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 Z1-Z56 Z1-Z56
Disable time measurements	NA	± 3.50 nS	Z1-Z56

2. The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

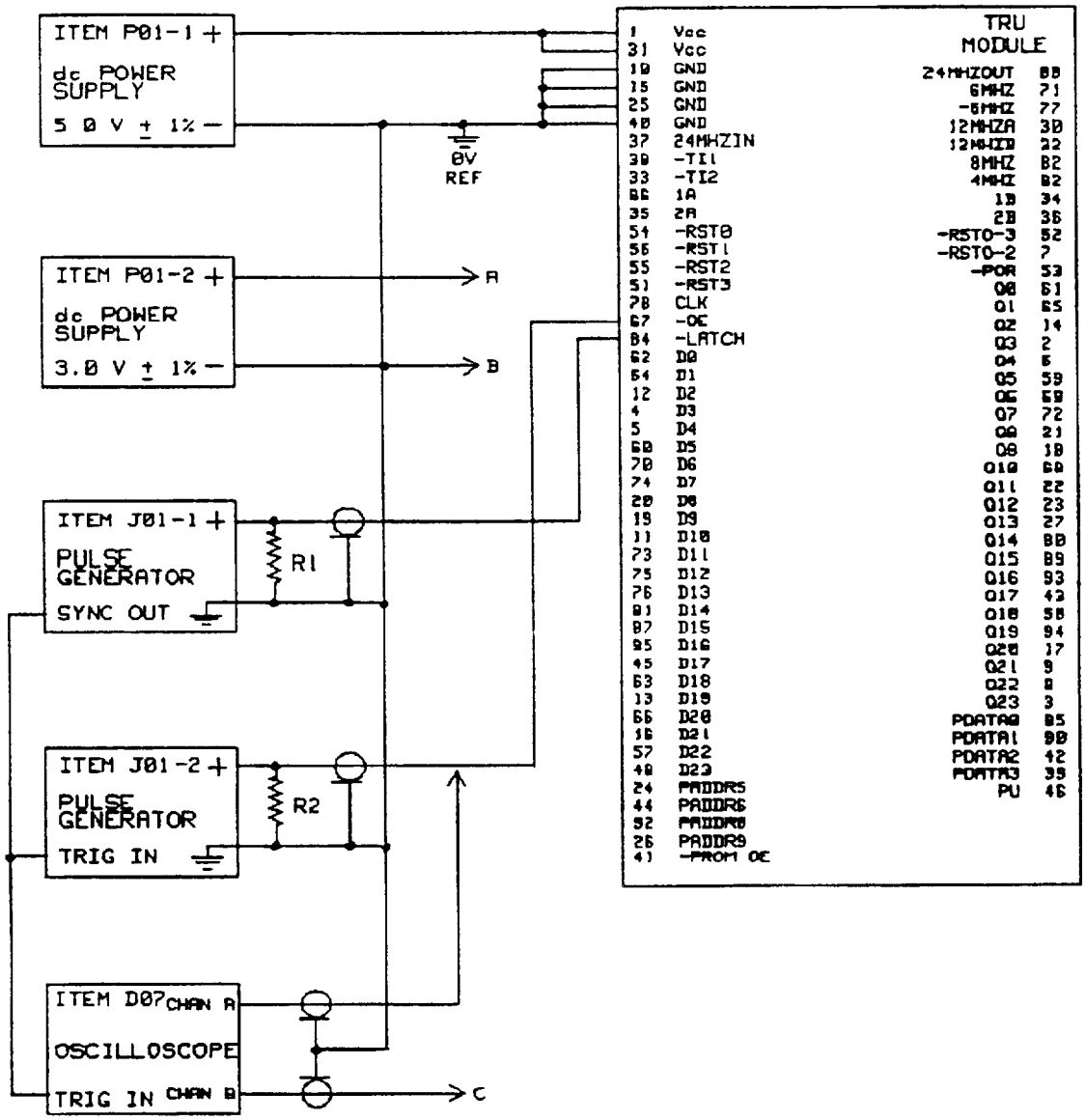
3. The test circuit shall be as indicated on figures 15 and 17.

4. Module inputs shall meet the timing requirements shown on figures 16 and 18.

5. Module outputs shall be loaded as specified on figure 7.

6. Abbreviations used are defined in 6.1.

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NOTE:

1. R1 = R2 = 51 Ohms ± 5 percent, 1/2 W.

FIGURE 15. Disable time test circuit for test conditions Z1-Z48.

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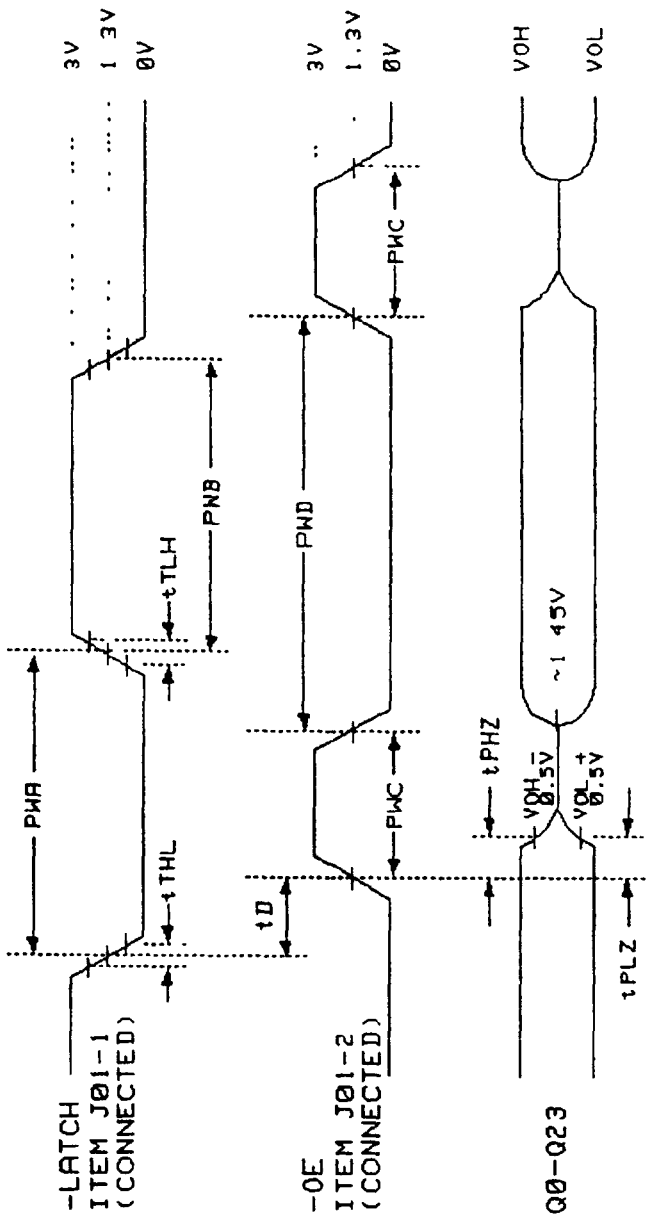


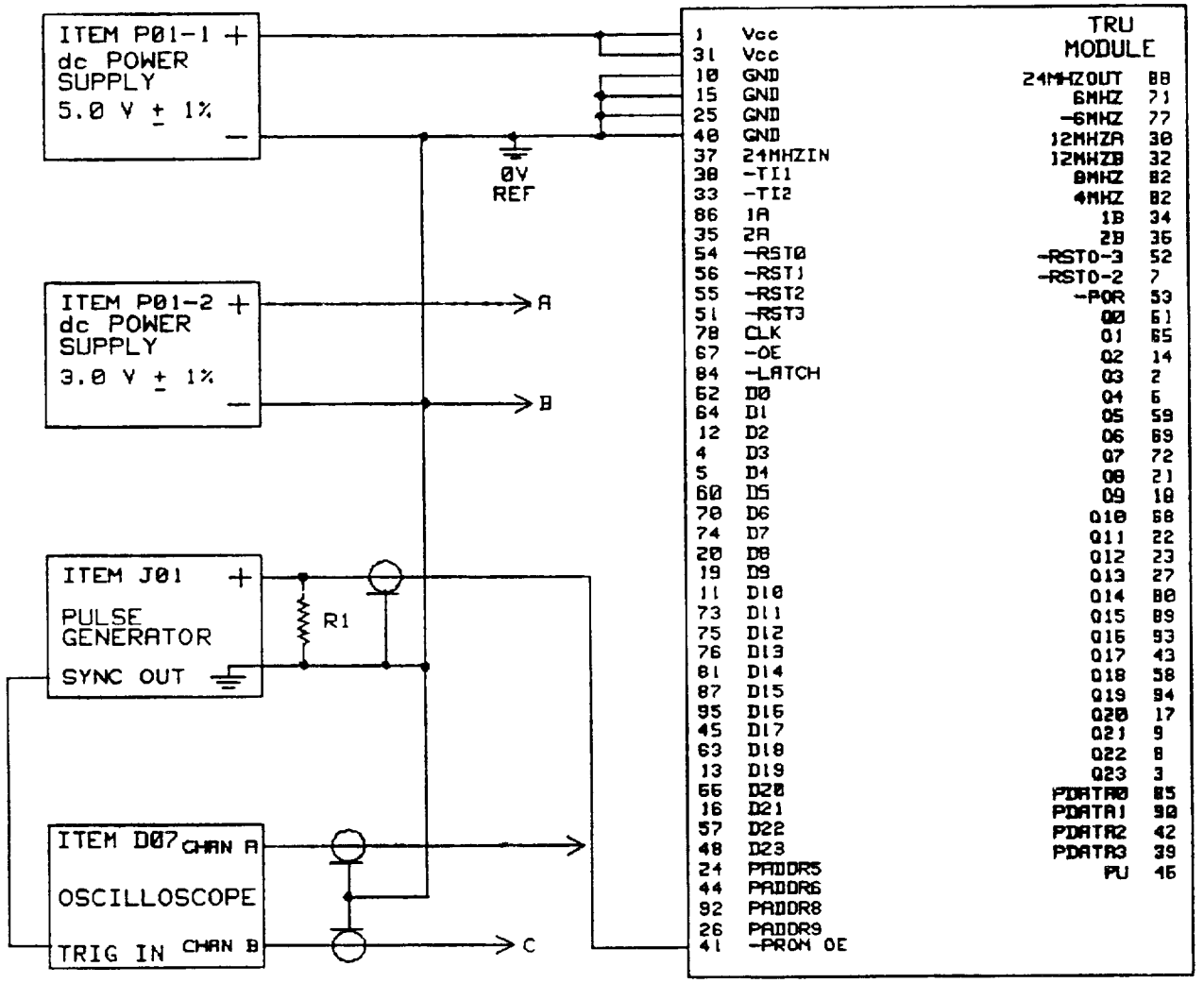
FIGURE 16. Disable time test waveforms for test conditions Z1-Z48.

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NOTES:

1. PWA = PWB = 500 nS.
2. PWC = 250 nS.
3. PWD = 750 nS.
4. tD = 125 nS.
5. tTHL = tTLH $\leq$ 10 nS.

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NOTE:

1. R1 = 51 Ohms ± 5 percent, 1/2 W.

FIGURE 17. Disable time test circuit for test conditions Z49-Z56.

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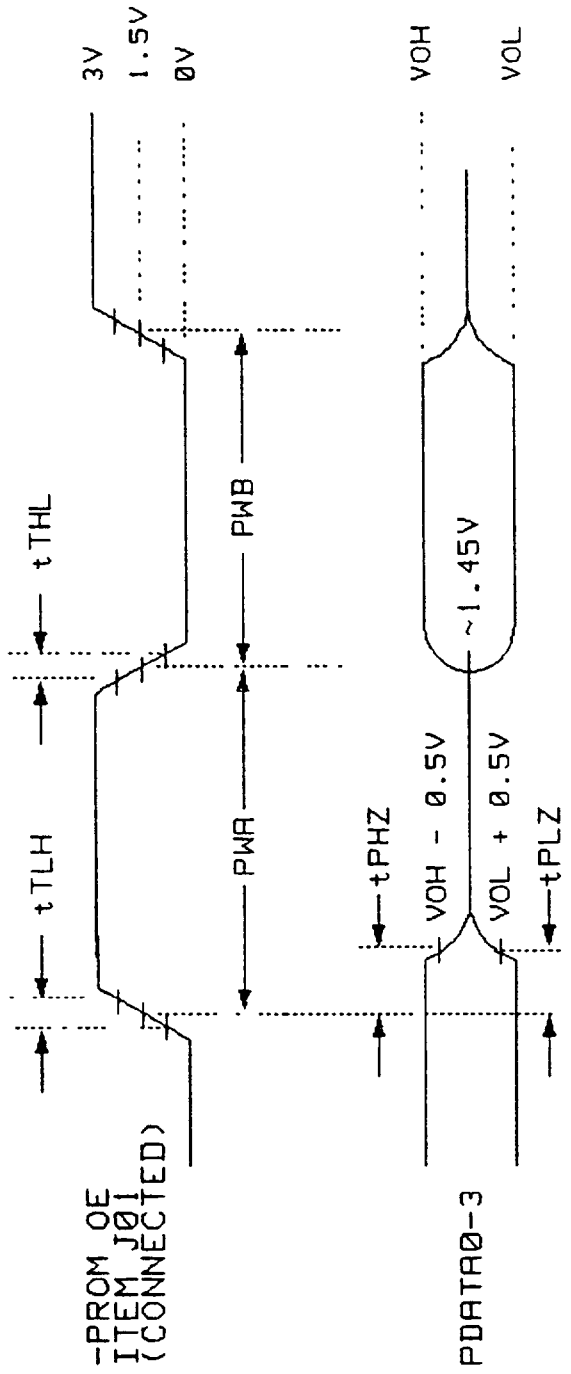


FIGURE 18. Disable time test waveforms for test conditions Z49-Z56.

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NOTES:

1. PWA = PWB = 250 nS.

2. tTHL = tTLH $\leq$ 10 nS.

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TABLE XXVII. Output pulse width and pulse period test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes	Fig	Ref
AA1	4MHZ PWL	29	117.0	133.0	nS	1-4	19	13
AA2	4MHZ PWH	29	117.0	133.0	nS	1-4	19	14
AA3	4MHZ PP	29	249.98	250.02	nS	1-4	19	15
AA4	6MHZ PWL	71	78.0	88.0	nS	1-4	19	7
AA5	6MHZ PWH	71	78.0	88.0	nS	1-4	19	8
AA6	6MHZ PP	71	166.65	166.68	nS	1-4	19	9
AA7	-6MHZ PWL	77	78.0	88.0	nS	1-4	19	10
AA8	-6MHZ PWH	77	78.0	88.0	nS	1-4	19	11
AA9	-6MHZ PP	77	166.65	166.68	nS	1-4	19	12
AA10	12MHZA PWL	30	35.0	48.0	nS	1-4	19	1
AA11	12MHZA PWH	30	35.0	48.0	nS	1-4	19	2
AA12	12MHZA PP	30	83.32	83.34	nS	1-4	19	3
AA13	12MHZB PWL	32	35.0	48.0	nS	1-4	19	4
AA14	12MHZB PWH	32	35.0	48.0	nS	1-4	19	5
AA15	12MHZB PP	32	83.32	83.34	nS	1-4	19	6

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NOTES:

1. Output pulse width and pulse period, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	AA1-AA15
Power supply current limit	0.966 A	NA	AA1-AA15
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	AA1-AA15
low, (VL)	0.0 V	± 100 mV	AA1-AA15
Pulse width and pulse period measurement	NA	± 0.5 nS	AA1,AA2
	NA	± 6.0 pS	AA3
	NA	± 0.5 nS	AA4,AA5
	NA	± 6.0 pS	AA6
	NA	± 0.5 nS	AA7,AA8
	NA	± 6.0 pS	AA9
	NA	± 0.5 nS	AA10,AA11
	NA	± 6.0 pS	AA12
	NA	± 0.5 nS	AA13,AA14
	NA	± 6.0 pS	AA15

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the point indicated on figure 19. The reference numbers correspond to the test conditions.

(c) Module pin 88 shall be connected to pin 37. The wire used for connection shall not exceed three (3) inches.

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NOTES: - Continued.

3. All outputs shall be loaded as specified on figure 7.

4. Abbreviations used are defined in 6.1.

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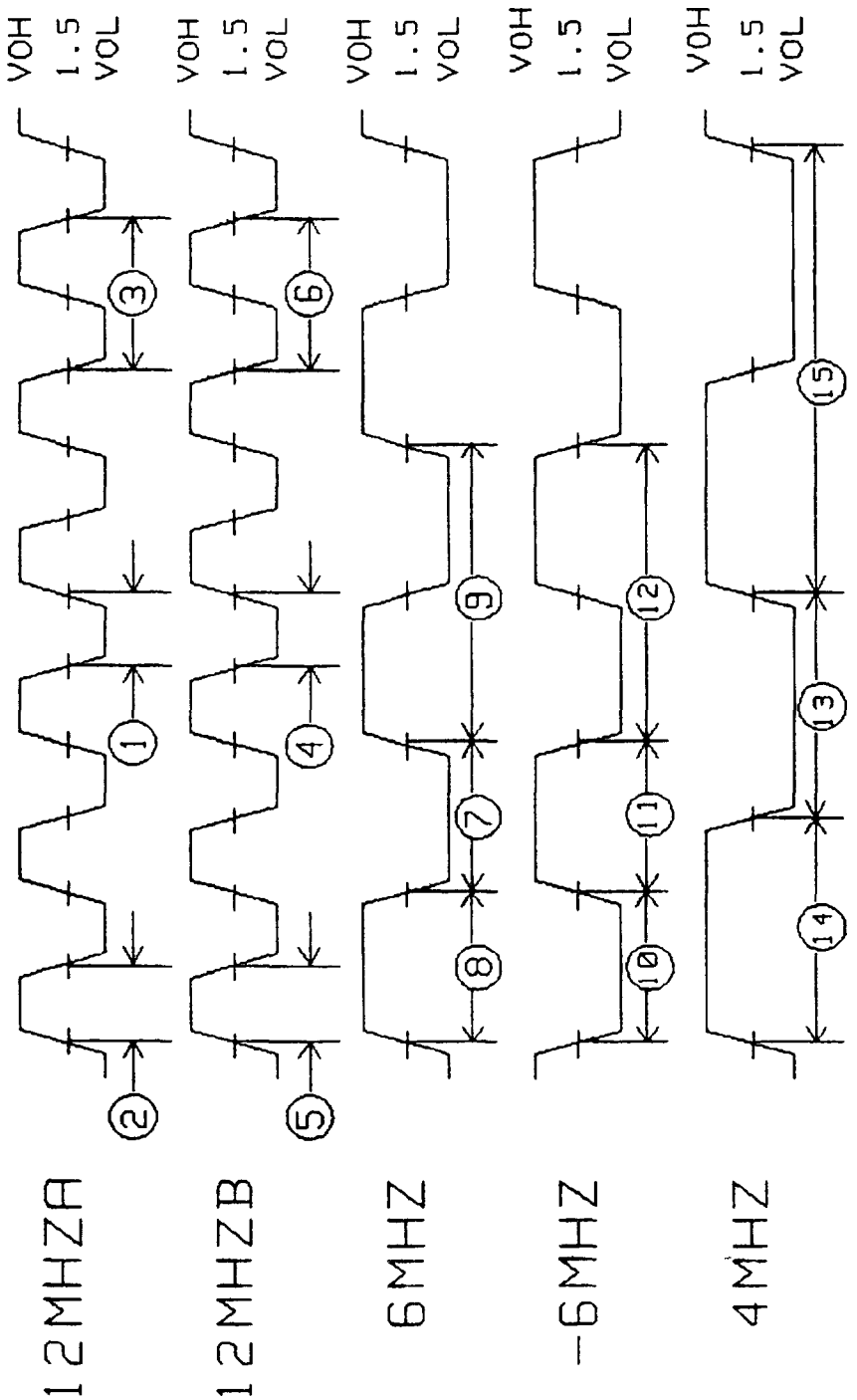


FIGURE 19. Output pulse width and pulse period test waveforms for table XXVII.

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	MIL-W-28787/347	1990 October 26

3 DOCUMENT TITLE

Modules, Digital, Standard Electronics, Clock, Microprocessor Module, Key Code TRU

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible Attach extra sheets as needed.)*

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